BISMILLAH RAHMAN RAHIIM
THE NATIONAL MUSLIM COUNCIL OF TANZANIA
ZAIM EDUCATION DEVELOPMENT LIMITED

FORM FOUR ZAIM SCHOOLS MOCK EXAMINATION

061 **BIOLOGY 2A (PRACTICAL)**
MARKING SCHEME

(For Both School and Private Candidates)

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1. (a) The aim of the experiment is to test the presence of starch in variegated leaf. **(4 MARKS).**

   (b) The leaf was boiled in water in order to kill the cells of a leaf (protoplasm). **(4%MARKS)**

   (c) (i) The leaf was boiled in ethanol in order to remove the green pigment of leaf (chlorophyll). **(4 MARKS)**

   (ii) To remove the ethanol. **(1 MARKS)**

   (d) After the addition of Iodine solution onto the leaf the expected results was the colour of leaf to change to blue-black. **(4 MARKS)**

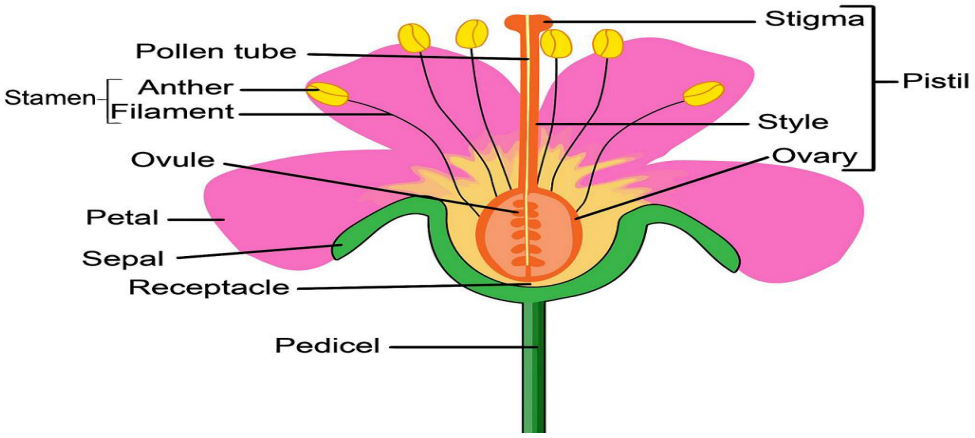
   (e) The leaf has been performing photosynthesis and producing starch. **(4 MARKS)**

   (f) Photosynthesis

   Photosynthesis is the process whereby green plants and other animals manufacture their food by using the carbondioxide, water and chlorophyll. **(4 MARKS)**

2. R – Bread mould
   S – Hibiscus flower **@ ($\frac{1}{2}$ Marks)**
   T – Bee **total (2 Marks)**
   U – Millipede

   DIAGRAM OF SPECIMENS



Title = **(1/2 MARKS)**
Drawing = **(1 MARKS)**
Labelling = **@ ½ = (3 1/2 MARKS)**

(iii) **Specimens**

Kingdom – plantae

Division – Angiospermophyta **(1/2 MARKS)**

Class - dicotyledonae

Specimen I

Kingdom – animalia

Phylum - arthropoda **(1/2 MARKS)**

Class – Insecta

Specimen U.

Kingdom - animalia **(1/2 MARKS)**

Phylum – arthropoda

Class – diplopoda

2. (iv)

SPECIMEN I		SPECIMEN U
(i)	The body is clearly divide into three region (Head, thorax and abdomen)	(ii) The body is divided into segments. (ii) They lack wings. (iii) They do not have a compound eyes. (iv) They have two pairs of jointed legs
(ii)	Most adult forms have wings.	
(iii)	They have pairs of compound	

eyes. (iv) They have three pair of jointed legs.	in each segments. **@ 1 MARKS** **TOTAL (4 MARKS)**

2. (b) (i) Economic Importance of specimen T are

i. They are used as the agent of pollination.

ii. They are used in biological study. **@ (1 MARKS)**

iii. They produce honey.

2. (ii) Distinctive features of specimen T are

i. They have pairs of antenae.

ii. They have pairs compound eye. **@ (1 MARKS)**

iii. They have three pairs of jointed legs.

iv. The body is divided into three region (head, thorax and abdomen).

2. (b) (iii) Kingdom fungi

Division Zygomycota **@ ($^{1}/_{2}$ MARKS)**

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FORM FOUR ZAIM SCHOOLS MOCK EXAMINATION

061 BOOK KEEPING

MARKING SCHEME

SECTION A

1.

i	ii	iiii	iv	v	vi	vii	viii	ix	x
C	B	B	D	D	B	D	B	E	C

2.

i	ii	iii	iv	v	vi	vii	viii	ix	x
J	A	E	H	C	I	B	G	D	F

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FORM FOUR ZAIM SCHOOLS MOCK EXAMINATION

CHEMISTRY
MARKING SCHEME

(For Both School and Private Candidates)

1.

I	II	III	IV	V	VI	VII	VIII	IX	X
B	A	D/E	D	A	A	B	D	A	D

@ 1MARK TOTAL 10MARKS

2.

I	II	III	IV	V	VI	VII	VIII	IX	X
H	F	E	O	D	J	A	L	M	N

@ 1MARK TOTAL 10MARKS

3. (a) (i) The electro negativity increase from A to C and decrease from B to D

(ii) the electronic configurations of A 2:8:8:2

C^{2-} 2:8:8

D 2:8:7

B 2:7

(b) (i) Basic oxide

(ii) $H + Cl \longrightarrow HCl$

(iii) $NaO_{aq} + HCl_{aq} \longrightarrow NaCl_s + H_2O_l$

4. (a) they expect higher cost because of

(i) hard water leaves a scummy deposited on clothing and utensil that are

Difficult to wash out

(ii) hard water need more soap than soft water.tis means a lot of soap wasted

(iii) The method used to remove it a so expensive

(iv) it cause lime scaling and blockage in various appliance

(b) (i) sublimation

(ii) grinding

(iii) fraction distillation

(iv) magnetization

5. (a) (i) R-OH alcohol

R-COOH carboxylic acid

RCOOR' ester

(ii) R-OH OH

R-COOH COOH

(b) (i) does not support combustion
Colourless
(ii) fire extinguisher

6. (a) solid fuel example **wood** fuel, coal fuel, charcoal fuel
Liquid fuel example kerosene, diesel, petrol
(b) (i) the number of moles in reactant side a higher than in the product side this
Cause when increase in pressure favors the forward reaction which produce More methanol.
(ii) the negative sign indicate exothermic reaction
(iii) since the reaction is exothermic lowering temperature favors the forward
Reaction that yield more production of methanol.

7. (a) (i) hydrogen peroxide(H_2O_2) and manganese (iv) oxide as catalyst its test is

Relight glowing splint

(ii) dilute hydrochloric acid (HCl) and zinc granules its test is pop sound

(iii) dilute hydrochloric acid (HCl) and calcium carbonate ($CaCO_3$) its test

Using lime water ($Ca(OH)_2$)

(b) (i) Methyl orange(MO) (ii) Any indicator (iii) phenolphthalein indicator (P.O.P)

8. (a) $2Hcl_{aq} + NaOH_{aq} \longrightarrow 2NaCl_{aq} + H_2O_L$

DATA GIVEN

MOLARITY OF ACID (MA) = 0.1M MOLARITY OF BASE MB ?

NUMBER OF MOLES OFBASE (Nb)= 1g/mol

VOLUME OF ACID (VA) = $25cm^3$ MOLAR MASS BASE 40 g/mol

VOLUME OF BASE (VB) = $23.00cm^3$

NUMBER OF MOLES OF ACID (Na)= 1g/mol

SOLUTION

$$\text{FROM } \frac{MA \text{ X } VA}{Na} = \frac{MB \text{ X } VB}{Nb}$$

$$MB = \frac{MAX\ VAX\ Nb}{Na \text{ X } VB} = \frac{0.1M \text{ X}25cm3 \text{ X } 1g/mol}{\frac{1g}{mol} \text{ x}23cm3}$$

MB = 0.1M

$$\text{But molarity} = \frac{\text{concentration g/dm3}}{\text{molar mass g/mol}}$$

concentration g/dm3 = molarity (mol/dm^3) x molarmass(g/mol)

0.1 x 40

concentration g/dm3 = 4 g/dm^3

(b) Soil structure. is the top layer of loose material covering the earth's

surface

Acidic soil. Is the type of soil which containing PH from 1-6

Liming is the application of calcium in various form of soil in order to

increase the PH of soil

9. (a) DATA GIVEN

CURRENT (I) = 0.5 A TIME= 30MINUTES = 1800SEC

MASS = ? EQUIVALENT WEIGHT ?

$$M = ZIT \text{ but } Z = \frac{\text{Mr of Ag}}{\text{Valency x faradays}}$$

$$M = \frac{\text{Mr of Ag}}{\text{Valency x faradays}} xIT$$

$$M = \frac{108}{1 \times 96500} x0.5X\ 1800 = 1.0g$$

$$\text{EQUIVALENT WEIGHT} = \frac{\text{Mr of Ag}}{\text{Charge of Ag}}$$

108/1

EQUIVALENT WEIGHT = 108

(b) Addition reaction is the one in which molecule is added to another

Example halogenations of alkene

Elimination reaction is the one in which molecule is added to another

Example alcohol react with conc sulphuric acid

10. (a) (i) nitrogen monoxide, nitrogen dioxide and chloro floro carbon (CFC)

(ii) (a) acid rain
(b) green house } with explanation

(c) global warming

(b) (i) provide education how to prevent water pollution

(ii) provide chemical in source of water which can clean and make safe for use

(iii) appropriate law should be put to control water pollution

11. (i) $Cs + O_2 \longrightarrow CO_2g$

(ii) the carbon dioxide formed react with more coke to form carbon monoxide

(ii) calcium oxide(solid) + silicon dioxide $\longrightarrow$ caliumsilicate

12. (i) any three hydrocarbon with four carbon (ALKANE, ALKENE,ALKYNE)

(ii) any four name of homologous series from one hydrocarbon with its structural formula

(iii)position and branched isomer

13. Any explained four point of important of manure in essay form

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FORM FOUR ZAIM SCHOOLS MOCK EXAMINATION

032/2 **CHEMISTRY 2A**
MARKING SCHEME
(For Both School and Private Candidates)

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1. (a) The colour of the solution change because of the presence of indicator

   (b) The average titre value 24.50$cm^3$for pipette of 25cm^3 and 19.60cm^3 for pipette of

   20cm^3

   (c) (i) 25cm^3 of solution ZZ required 24.50cm^3 of solution MM for complete reaction

   (ii) 20cm^3 0f solution ZZ required 19.60r cm^3 of solution MM for complete reaction

   (d) the titre value will be difference because of different number of moles. In hydrochloric acid vs sodium carbonate the ration is 1:2(Nb=1,Na=2) but when sulphuric acid used the ratio will be 1:1(Nb=1,Na=1).

   (e) $2Hcl_{aq} + Na_2CO_{3aq} \longrightarrow 2NaCl_{aq} + H_2O_L + CO_{2g}$

   (f) (i) DATA GIVEN

MOLARITY OF MM = 0.1M NUMBER OF MOLES OF MM= 1g/mol

VOLUME OF ZZ = $25cm^3$ MOLARITY OF ZZ ?

VOLUME OF MM = $24.50cm^3$ MASS OF IMPURE ZZ = 1.37g

NUMBEROFMOLEOF ZZ= 2g/mol VOLUME OF IMPURE $250cm^3$=$0.25dm^3$

SOLUTION

FROM $\frac{M(MM)V(MM)}{n(M)} = \frac{M(ZZ)V(ZZ)}{n(Z)}$

$$M(ZZ) = \frac{M(MM)X\ V(MM)X\ n(Z)}{n(M)X\ V(ZZ)} = \frac{0.1M\ X24.50cm3\ X\ 1g/mol}{2g/mol\ x25cm3}$$

M(ZZ) = 0.049M

Conc impure = 1.37/0.25

Conc impure = $5.48g/dm^3$

Conc pure = molarity x molar mass

0.049 x 106= $5.194g/\ dm^3$

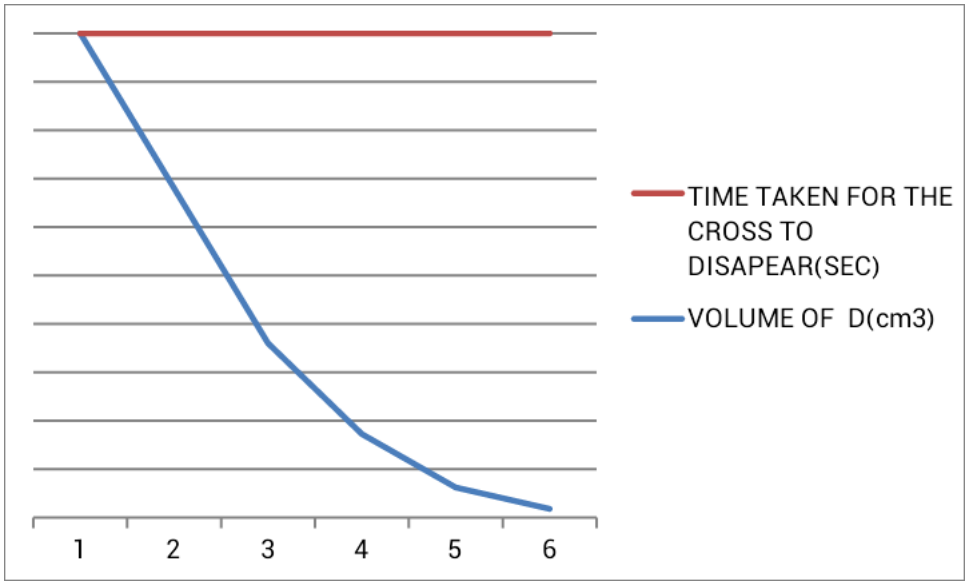
$$\%purity = \frac{pure\ conc}{impure\ conc} x\ 100$$

$$\frac{5.194}{5.48} x\ 100 = 94.78\%$$

2. (a)

VOLUME OF D(cm^3)	VOLUME OF S(cm^3)	TIME TAKEN FOR THE CROSS TO DISAPEAR(SEC)	$\frac{1}{T}$ sec^{-1}
30	0	0	0
25	5	12	0.083
20	10	36	0.027
15	15	72	0.013
10	20	152	0.006
5	25	280	0.003

(b) (i)



(b) (ii)the graph indicate that the rate of reaction is slow at low concentration of sodium thiosulphate and faster at higher concentration.

(c) $Na_2S_2O_{3aq} + Hcl_{aq} \longrightarrow NaCl_{aq} + S_s + H_2O_l + SO_{2g}$

(d) the cross disappear due to the formation of sulphur

3.

S/N	Experiments	Observation	Inference
(a)	Observe sample K	Sample K was deep blue crystalline	Cu^{+2} may be present
(b)	Put a spatulaful of sample K in attest tube and add distilled water	Sample K dissolved in distilled water and form blue solution	Cu^{+2} may be present
(c)	transfer 0.5g of sample K in a test tube , add dilute HCL	A colour less gas with a chocking smell	SO_3^{-2} or SO_4^{-2} may be present
(d)	Transfer 0.5g of sample K in a test tube , add concentrated sulphuric acid.	No gas produced	SO_4^{-2} may be present
(e)	Dissolve sample K then divide the resulting solution into three portion i. To the first portion add sodium hydroxide solution	Blue precipitate insoluble in excess,the ppt turn black on boiling	Cu^{+2} present
	ii. To the second portion add ammonia solution	Pale blue ppt soluble in excess forming deep blue solution was observed	Cu^{+2} present and confirmed
	iii. To the third portion add potassium ferrocyanide solution	Redish brow ppt formed	Cu^{+2} confirmed
	iv. To the furt portion add barium chloride followed by hydrochloric acid	Immediately white ppt insoluble in HCl was observed	SO_4^{-2} confirmed

Conclusion

A. (i) The cation in sample K was Cu^{+2}

(ii) The anion in sample K was SO_4^{-2}

(ii) The compound K was $CuSO_4$

B. $CuSO_{4aq} + Hcl_{aq} \longrightarrow CuCl_s + H_2SO_{4aq}$

Net ionic equation

$Cu^{+2}{}_{aq} + Cl^-{}_{aq} \longrightarrow CuCl_s$

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FORM FOUR ZAIM SCHOOLS MOCK EXAMINATION

011 **CIVICS**
MARKING SCHEME
(For Both School and Private Candidates)

SECTION A (20 Marks)

1. Multiple choices. **(@1x10=10 marks)**

Question	I	ii	iii	Iv	V	vi	vii	viii	Ix	x
Answer	B	E	A	C	A	B	D	B	C	E

2. Matching items. **@1x10=10 marks**

List A	I	ii	iii	Iv	V	vi	vii	viii	ix	x
List B	H	F	J	B	A	R	Q	P	M	D

SECTION B (20 Marks)

3. (a) Title of the passage: ECONOMIC ACTIVITIES **(02 marks)**

   (b) Sources of labour in Tanzania **(4 points @0.5=2 marks)**

   i. Peasants

   ii. Small scale farmers

   iii. Livestock keepers

   iv. Traders

   v. Workers in crafts

   vi. Industrial workers

   (c) The way Tanzania benefit from livestock keeping sector **(@1x2=2 marks)**

   i. Livestock keepers provide Tanzania with meat, milk and eggs which are very important as a source of protein.

   ii. Livestock keeping is a source of government revenue.

   (d) Problems that hinder fishing industry in Tanzania **(@1x2=2 marks)**

   i. Lack of expert

   ii. Lack of efficiency modern fishing methods.

   (e) Ways of improving agriculture in Tanzania **(4 points @0.5=4 marks)**

   i. Education to the farmers
   ii. Application of modern technology eg. Tractors and high breed seeds.
   iii. Using of various pesticides and insecticides to control pests and diseases.
   iv. Improvement of storage facilities.

TOTAL 10 MARKS

4. (a) (i) Local government refers to the government which formed at the grass-root level in order to assist the central government in matters of administration and other activities. **(1Marks)**

   (ii) Sources of local government revenue **(4 points @1=4 Marks)**

   (a) Rent
   (b) Licenses
   (c) Duties
   (d) Contributions
   (e) Grants and loans from Central government

   (b) Importance of National Constitution **(5 points@1=5)**

   i. The constitution protects the rights of individuals.

   ii. It describes the form of government that is used in the country.

   iii. It defines the limits of the power of the leaders.

   iv. It shows how leaders are to be chosen

TOATAL 10 Marks

SECTON C (60 Marks)

5. Improper behavior is a behavior which is not accepted in the society. Example laziness, prostitution using of abusive language, drug abuse among other. **(01 Marks)**

   - Causes of improper behavior;
     - Drunkenness
     - Poverty
     - Lack of proper education
     - Lack of good parental care and guidance
     - Mass media
     - Peer-pressure
   - Any **six (6)** relevant points **@3 Marks=18 Marks**
   - Relevant conclusion **(01 Marks)**

TOTAL 20 Marks

6. Globalization is the process of increasing the connectivity and interdependence of the world's economy, political, culture, science and technology among the people or nations. (01 Marks)

   - Positive impacts of globalization;
     - Effective utilization of natural resources
     - Open door for employment opportunities
     - Increase of flow of product to our markets
     - Improvement of quality of goods and services
     - Increase in the spread of information and public awareness
     - Increased the spread of international language
   - Any **six (6)** relevant points **@3 Marks=18 Marks**
   - Relevant conclusion **(01 Marks)**

TOTAL 20 Marks

7. Private sector is that category of the country in which production of goods and serves are not owned and controlled by the government. (01 Marks)

   - Advantages of private sector to the Tanzania economy;
     - Help to raise government revenue
     - Help to increase employment opportunities
     - Leads to technological advancement
     - Promote competition leading to greater efficiency in production
     - Help to reduce the government burden in providing services
     - Increase the rate of investments in the country
   - Any **six (6)** relevant points **@3 Marks=18 Marks**
   - Relevant conclusion **(01 Marks)**

TOTAL 20 Marks

8. Life skills are the abilities that a person has to live well in the society. OR

   Are the mental knowledge and ability which enable human to manage or control him/herself and able to act, think or behave in acceptable manner within a community. **(01 Marks)**

   - Consequences of not applying life skills;
     - It may lead to occurrence of conflicts
     - It cause poor decisions
     - Increase of poverty
     - It may lead a person to end in bad groups
     - It may lead to bud behavior
     - Failure to create health relationship
   - Any **six (6)** relevant points **@3 Marks=18 Marks**
   - Relevant conclusion **(01 Marks)**

TOTAL 20 Marks

9. Democracy is the system of government in which people rule. **OR**

   Is the system of government in which all people of a country can vote to elect their representatives. **(01 Marks)**

   - Democracy is promoted in Tanzania through the presence of the following factors;
     - Regular free and fair election
     - The presence of rule of law
     - Transparent and accountability
     - Economic freedom
     - Equality
     - Citizen participation
     - Presence of multiparty system

- Any **six (6)** relevant points **@3 Marks=18 Marks**
- Relevant conclusion **(01 Marks)**

TOTAL 20 Marks

10. Road accidents are sudden unexpected incidents which occur in the road and cause injury, loss of life or destruction of properties. **(01 Marks)**

- Causes of road accidents (caused by man);
    - Drunk driving
    - Overworking
    - Reckless driving
    - Day-workers
    - Pedestrians do not obey road safety rules
    - Overload of vehicles
- Any **six (6)** relevant points **@3 Marks=18 Marks**
- Relevant conclusion **(01 Marks)**

TOTAL 20 Marks

061 **COMMERCE**

(For Both School and Private Candidates)

MARKING SCHEME

Time: 2:30 Hours **Tuesday, 2nd June 2019 a.m**

SECTION A

1. **(1 MARK EACH)**

i	ii	iii	iv	v	vi	vii	viii	ix	x
C	B	E	B	C	C	C	A	C	A

2. **(1 MARK EACH)**

i	ii	iii	iv	v	vi	vii	viii	ix	x
Q	S	J	M	B	O	D	E	H	K

SECTION B (10 MARKS, 2 MARKS EACH)

3. **(A)** **DOUBLE INSURANCE** is an insurance whereby more than one policy is taken to cover the same risk but the insured cannot be recovered more than insured value of the properly from his insurers.

   (B) **RE – INSURANCE** is a procedure under which an insurance company enters into a contract with another insurance company with assumption that the insurer solely cannot compensate wholely insured properly due to high value.

   (C) **COVER NOTE** is the temporary insurance contract act as evidence of agreement to cover any loss that may happen between the time insurers accept the proposal and the time insurer issue the policy to the proposer.

   (D) **ASSESSORS** these are insurance officers who calculate the amount of danger involved in any risk when the insured makes a claim on the insurer.

   (E) **PREMIUM** is the amount of money the insured pays to the insurer as the consideration to the latter's undertaking to compensate him in event of loss.

4. **(A)** **FEATURES OF A GOOD TAX SYSTEM (1 Mark Each)**
   i. Certainty
   ii. Convenient
   iii. Economy

iv. Productivity
v. Equity

(B) MERITS OF DIRECT TAX (1 Mark Each)

i. Low cost of collection.
ii. Tax payer know how much to pay.
iii. It encourage a tax payer to pay since he/she knows how much to pay.
iv. It brings equality because high incomes pay high tax while low income pays low taxes.
v. It is difficult to avoid.

5. **DIFFERENCE OF THE FOLLOWING TERMS (2.5 Marks Each)**

(A) Direct production is the method of production whereby goods and services are produced by an individual for his/her own use.

WHILE

Indirect production is the method of production whereby goods and services are produced by an individual for the sake of exchange.

(B) Chain store is the large scale retail shops which are scattered in different geographical area with one ownership and management.,

WHILE

Department store is number of shops in a single building selling different types of goods owned by one individual or firm.

(C) Bill of loading is a document issued by a shipmaster or ship owner to the importer contain details of goods carried in a ship.

WHILE

Charter party is a document which contain agreement made by shipper and ship owner for transporting imported or exported goods.

(D) Share is the part of capital of the company.

WHILE

Debenture is the money which a company borrows from the public.

6. (A) $P_0 = 40/=$ $Q_0 = 800$ ***(2 MARKS)***
$P_1 = 20/=$ $Q_1 = 1000$

$$\text{Price elasticity of demand} = \frac{\text{Percentage change in Q}}{\text{Percentage change in price}}$$

$$\text{Percentage change in Quantity} = \frac{Q_{1-Q_0}}{Q_0}$$

Percentage change in Price = $\frac{P_{1-P_0}}{P_0}$

Price elasticity or demand = = $\frac{\frac{Q_{1-Q_0}}{Q_0}}{\frac{P_{1-P_0}}{P_0}}$ ***(3 Marks)***

$$= \frac{\frac{1000-800}{800} \times 100}{\frac{20-40}{40} \times 100} =$$

$$= \frac{25\%}{50\%} = -0.5$$

Price elasticity of demand = -0.5. (2 Marks)

(B) It is elastic because the proportionate change in quantity demanded (25%) is lower than the percentage change in price (50%). (3 Marks)

SECTION C

7. (A) **FEATURES OF PRIVATE LIMITED COMPANY. (1 Mark Each)**
   - Member's liability is limited to their capital.
   - Shares cannot be transferred.
   - The business is separated legal entity.
   - There is assured continuity.
   - Number of members is 2 to 50 members.

   (B) **Partnership deed** is a written agreement prepared by members who wish to start partnership business, it contains terms and conditions made between partners. ***(2.5 Marks)***

   CONTENTS OF PARTNERSHIP DEED (0.5 Marks Each)

   (i) Name, address and occupation of each partner.
   (ii) Rights and duties of each partner.
   (iii) Salaries to be paid to partner if any.
   (iv) Rate of interest to be paid on capital, drawings or loans.
   (v) Ratio of profit or loss to be shared.

8. **SERVICES OFFERED BY TANZANIA POSTAL OFFICE**
   - Introduction (3 Marks)
   - Services offered (2.5 Marks Each)

(i) Ordinary post – sending and receiving letters and parcels.
(ii) Money order.
(iii) Postal order.
(iv) Sending and receiving cheque.
(v) Post restate.
(vi) Business reply services.

- Conclusion (2 Marks)

9. Introductions. (3 Marks)
- Current trends in Tanzania banking and financial services. (3 Marks Each)
  (i) Modern mechanization i.e use of ATM machine.
  (ii) Online loan provision e.g TALA, BRANCH, M-PAWA.
  (iii) E – money e.g M- pesa, simbanking.
  (iv) Computerized banking management system.
  (v) Issue of credit and Debit cards.
- Conclusion. (2 Marks).

10. (A) Containerization is the process of parking the cargo in standardized containers
which are sealed by the exporter or his agents. (2 Marks Each)

ADVANTAGE OF CONTAINERIZATION. (3 Marks Each)
(i) Facilitate use of modern cargo-handling machine.
(ii) Saves time and labour handling cost.
(iii) Efficient use of space.
(iv) Safety of goods.
(v) Transporting containers is easy compared to transporting goods with irregular shape.

DISADVANTAGES OF CONTAINERIZATION. (3 Marks Each)
(i) Not suitable for small quantities of goods which are not enough to fill the whole container.
(ii) Require special machinery.
(iii) Increase transportation cost.

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ZAIM EDUCATION DEVELOPMENT

FORM FOUR ZAIM SCHOOLS MOCK EXAMINATION

ELIMU YA DINI YA KIISLAMU 1
(Kwa watahiniwa Walioko na Wasiokuwa Shuleni)

1. @ alama 1

i	ii	iii	iv	v	vi	vii	viii	ix	x
E	C	B	C	B	E	C	E	C	C

2. @ alama 1

i	ii	iii	iv	v	vi	vii	viii	ix	x	xi	xii	xiii	xiv	xv
G	I	J	F	Q	D	A	C	K	N	P	U	B	R	S

3. @ alama 1

   A. III, V, I, IV, II.

   B. V, II, IV, I, III.

   C. IV, II, III, I, V.

4. @ $\frac{1}{2}$ Nukta nne (04)

   (a) (i) Kutuliza mwili na fikra

(ii) Kuzingatia juu ya ukubwa wa Allah

(iii) Kuelewe maana ya maneno ndani ya swala

(iv) Kuwa na hofu na adhabu ya Allah

(v) Kuwa na haya kutenda makosa

(vi) Kuwa na matumaini na rehema za Allah

(b) @ nukta 1

(i) Wasafiri

(ii) Wagonjwa

(c) nukta 1

(i) Ijab

(ii)Qabuli – kabuli

(d) @ $^{1}/_{2}$

i. Qur'an
ii. Sunna / Hadithi
iii. Ihtihad

(e) @ $^{1}/_{2}$

i. Harufu yake
ii. Ladha yake
iii. Rangi yake

(f) @ $^{1}/_{2}$

i. Josho mtu akiingia kaika Uislamamu

ii. Josho la siku ya Iddi

iii. Josho la siku ya Ijumaa

(g) @ $^{1}/_{2}$ nukta (4)

i. Tak-bira ya kuhirimia swala

ii. Kusoma suratul Faat-ha

iii. Kusmswalia Mtume sala na salamu juu yake

iv. Kusoma Tahahhudi

v. Kutoa salamu

(h) @ alama 3

- Talaka khuhui ni mwanamke kujivua kwenye ndoa kwa kumrejeshea mme mahari yake na atakaa Eda ya kuachwa ya kawaida

(i) @ alama 2

- Anakaa siku mia moja na thelathini (130)

(j) @ alama 2

- Ni vipande vitano vya nguo au mikaka mitabno. Kanzu mbili, shungi, gaguri na wikaya.

(k) @ $^{1}/_{2}$

i. Kilichoibiwa kiwe kimehifadhiwa vizuri

ii. Kiwe kimefikia robo dinari

iii. Ushahidi wa wizi ukamilike

(l) @ alama 2

- Hukumu yake ni kuuawa kwa mujibu wa sheria ya kiislamu “Mwenye kuacha dini muuweni”

(m) @ $^{1}/_{2}$

i. Ngamia watano mbuzi (1)
ii. Ngamia kumi mbuzi (2)
iii. Ngamia kuminatano mbuzi (3)
iv. Ngamia Ishirini Mbuzi (4)
v. Ngamia Ishrini natano Ngamia mmoja
vi. Mbuzi arobaini (40) mbuzi mmoja

(n) @ $^{1}/_{2}$

i. Kunyoa
ii. Kutufu
iii. Kupiga jamarati
iv. Kuchinja

(o) Alama 02

Mali ya kuokota nay a kuchimbuliwa ardhini hazina Nisaab bali hutolewa $^{1}/_{5}$ ya thamani ya mali yote

5. @ Alama 30

a) Asmaa Allah amridhiye, Ghaari thauri
b) Uhud, Abu J agal
c) Umar Bin khatab Allah amridhiye, lulu Al-majusi (fizad)
d) Mtume , Mtu
e) Haruna Alaihi salamu, Ufasaha
f) Hudaibiya, Othmani bin Affan Allah amridhiye
g) J abal safaa, Abu lahab
h) Warumi na Wajemi, Othman bin Affan Allah amridhiye

i) Abdul-rahman bin Muljam, Baraka bin Abdallah

j) Yusufu Alaih salam; Joseph

k) 309, Yusufu Alaih salaam

l) Suleyman Alaih Salaam, Daud Alaih salaamu

m) Yathrib, Abu Ayoub Al-Answar Allah amridhiye

n) Qur'an, alama za kuwepo Allah Azza wajallah

o) Adam na Hawa Alaihima salaam, pepono

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015/1 ELIMU YA DINI YA KIISLAMU 2
(Kwa watahiniwa Walioko na Wasiokuwa Shuleni)

SEHEMU A(QURAN) Alama (25)

(1)Mgawanyo: Utangulizi alama **(1.5)**

Tafsiri ya sura ,alama**(4)** kila ayajumla ni **12**

Mafunzo alama alama**(2**)kila funzo jumla ni **10**

Hitimisho alama **(1.5)**

JUMLA 25

1. Hakika tumekupa kheri nyingi.
2. Basi swali kwa ajili ya Mola wako na uchinje (kwa ajili yake pia).
3. Hakika adui yako ndiye atakayekuwa mkiwa (atakatikiwa na kila kheri).

Mafunzo ya Sura

1. Mafanikio hayapimwi kwa wingi wa mali na watoto, bali ni kuridhiwa na Mwenyezi Mungu (s.w).
2. Hakuna neema kubwa kama kujaaliwa kuupokea Uislamu, kuuelewa, kuufuata vilivyo na kuufikisha kwa wengine.
3. Hatuna budi kumuabudu Mwenyezi Mungu (s.w) peke yake ipasavyo na

kufanya kila jambo kwa ajili yake tu.

4. Maadui na Wapinzani wa Uislamu ndiyo wenye kupata khasara na kufedheheka duniani na Akhera pia.
5. Ushindi na mafanikio hupatikana kwa waislamu baada ya kutii na kufuata maamrisho ya Uislamu vilivyo.

(2) - Mtume (s.a.w) aliwaamrisha maswahaba wake kujifunza na kuifahamu Qur'an na kuihifadhi moyoni

- Mtume (s.a.w) aliteuwa baadhi ya maswahaba kuiandika Qur'an ilipokuwa inashuka.
- Kuhifadhiwa kwa kusomwa mara kwa mara.
- Qur'an pia ilihifadhika kwa kusomwa mara kwa mara katika ibada za swala.

Mgawanyo wa alama
Utangulizi alama(2.5)
Kila hoja alama (5) jumla 20
Hitimisho alama (2.5)
Jumla alama 25

(3) **(i)** Wanafuata baadh ya amri na kuacha baadhi
(ii) kutokudumu kwa sharia za Quran
(iii)Hawatumii Quran kama ni kinga yao
(iv) Hawasomi Quran
(v) wanapinga na kukataa amri za Allah (s.w)mfano mirath,ukewenza
(vi)Hawazingatii mafundisho ya Quran
(vii)Hawajui lengo la Quran

Utangulizi alama(2.5)
Kila hoja alama (5) jumla 20
Hitimisho alama (2.5)
Jumla alama 25

(4) SEHEMU B (TAWHEED)

i kuwapenda mitume kuliko nafsi zetu Quran (9:24)
ii Ni kuwatii kwa kuwafuta maamrisho na kuacha makatazo yao katika kuendea kila kila kipengele cha maisha kibinafsi ,kifamilia na kijamii(4:59)
iii Ni kuwaiga mienendo na tabia zao katika maisha ya kibinafsi,kiamii na kifamilia Quran(33:21)
ivNi kuwafanya Mitume kuwa Mahakimu wetu juu ya jambo lolote tunapohitilifiana. Rejea Qur'an (4:65), (33:36)
v. Ni kuyakinisha kuwa Mtume Muhammad (s.a.w) ndiye Mtume wa mwisho. Rejea Qur'an (33:40)

vi. Ni kumswalia Mtume Muhammad (s.a.w) mara kwa mara na kila anapotajwa. Rejea Qur'an (33:56

Mgawanyo wa alama
Utangulizi alama (**2)**
Kila hoja alama (**3.5)**jumla**21**
Hitimisho alama(**2)**
Jumla ni alama 25

(5) Ufufuo ni jambo losiloepukika mbele ya Allah (s.w)

iKufufuka kwa watu wa Nabii Musa (a.s) waliokufa walipotaka kumuona Allah (s.w) kwa dhahiri (jahara).Rejea Qur'an (2:55-56)
ii Kufufuka kwa mtu (Myahudi) aliyeuliwa kwa dhuluma zama za Nabii Musa (a.s) ili amtaje aliyemuua baada ya kupigwa na sehemu ya nyama ya ng'ombe.Rejea Qur'an (2:72-73)
iii. Nabii Isa (a.s) kufufua wafu (watu waliokufa) kwa idhini ya Allah (s.w) kama sehemu ya muujiza wake. Rejea Qur'an (3:49), (5:111)
iv Mtu (msafiri) na punda wake waliokufa walifufuka baada ya miaka mia (100). Rejea Qur'an (2:259)
v. Kufufuka ndege wanne (4) wa Nabii Ibrarahim (a.s) kama mujiza wa kuonesha uwezekano wa kufufuliwa viumbe. Rejea Qur'an (2:260

vi. Kuamka vijana wa pangoni baada ya kulala kipindi cha muda wa miaka 309 ili kuwa ushahidi wa kufufuka viumbe baada ya kufa. Rejea Qur'an (18:25)

Mgawanyo wa alama
Utangulizi alama (2)
Kiini alama (21)
Hitimisho alama (2)
J UMLA 25

(6) Sababu ya Uislamu Kuwa Ndiyo Dini Sahihi Pekee.

Uislamu ndio dini pekee inayostahiki kuchaguliwa na kufuatwa kama muongozo sahihi wa maisha na watu wote kwa sababu zifuatazo;

(a) Uislamu unaeleza ukweli juu ya maumbile na nafasi ya mwanaadamu hapa Duniani;

Uislamu unaeleza lengo la kuumbwa mwanaadamu kuwa ni ibada na kumtumikia Mwenyezi Mungu na kuwa msimamizi wa sheria na hukumu zake hapa ulimwenguni ili kupata radhi zake

(b) Sheria na kanuni za Uislamu zinaendana na kanuni za kimaumbile;

Uislamu ndiyo dini pekee inaenda na kuzingatia kanuni na taratibu zote za kimaumbile za kibailojia kifizikia na kikemia ya viumbe vyote vilivyo na visivyo hai ikiwa ni pamoja na maumbile au mazingira yanayotuzunguka.

(c) Uislamu ndio dini inayotosheleza matashi yote ya mwanaadamu kimwili na kiroho;

Uislamu ni dini inayotoa muongozo namna ya mtu kumiliki hisia zake na kuendea mambo katika hali tofauti tofauti kama vile wakati wa furaha, shida, huzuni, misiba, n.k.

(d) Uislamu ni njia kamili ya maisha;

Ni Uislamu pekee ndio mfumo kamili wa maisha ya kila siku ya mwanaadamu katika fani na nyanja zote za kibinafsi na kijamii kama vile kisiasa, kiuchumi na kiutamaduni.

(e) Uislamu una ibada maalum za kumjenga mwanaadamu kimaadili;

Uislamu umeweka muongozo katika kumuadilisha mwanaadamu kupitia ibada maalum kama vile swala, funga, hija, kutoa zakat na sadaqat na zinginezo za sunnah.

(f) Uislamu ni dini ya walimwengu wote na wa nyakati zote;

Uislamu umekuwa ni mwongozo wa maisha tangu binaadamu wa kwanza hadi wa mwisho katika Nyanja zote za maisha bila kujali taifa, ukoo, rangi na hadhi ya watu.

(g) Uislamu ni dini ya kusimamia haki, uadilifu na usawa;

Ni dini ya Uislamu ndiyo inayotoa na kusimamia kwa usawa na uadilifu haki, hukumu na adhabu zote kwa watu wote bila ya upendeleo na ubaguzi wa rangi, taifa, hadhi na ukoo.

Mgawanyo wa alama
Utangulizi alama (2)
Kiini alama (21)
Hitimisho alama (2)
J UMLA 25

SEHEMU C FIQHI

(7) Matendo ya ibada ya Hijjah na Mafunzo yatokanayo.

Kabla na wakati wa kutekeleza ibada ya Hijjah, kuna mafunzo na hatua za kuzingatia kama ifuatavyo

i. Maandalizi na Safari ya Hijjah.

Ni lazima muislamu kabla ya kwenda Hijjah afanye maandalizi

ikiwa ni pamoja na kuzidisha ucha-Mungu, kuandaa gharama, mahitaji,

ii. Ihram.

Ni hali ya kuwa katika vazi rasmi na miiko kwa mwanaume mwenye Kuhiji au kufanya Umrah

- Vazi la Ihram linatufunza usawa baina ya wanaadamu wote.

iii. Talbiya.

Ni kuitikia wito wa Mwenyezi Mungu (s.w) kwa kusema yafuatayo, *"Labbayka Allahumma Labbayka. Labbayka laashariika laka labbayka. Innal-hamda Wanni'imata laka wal-mulka, laashariika laka"*

- Talbiya inatufunza kumuitikia Mwenyezi Mungu (s.w) kufuata maamrisho na kuacha makatazo yake.
- Talbiya inatufunza umoja na usawa baina ya waislamu wote.

iv. Tawaf.

Ni kitendo cha kuizunguka Ka'aba mara saba kuanzia jiwe jeusi kwa mwelekeo wa kinyume na saa kwa kulibusu au kuligusa au kuashiria.

Mafunzo:

- Kuchukua ahadi ya utii kwa Mwenyezi Mungu (s.w).
- Kumtaja na kumtukuza Mwenyezi Mungu (s.w) kwa wingi.
- Inatufunza umoja, udugu na usawa baina ya wanaadamu wote.

v. Sai.

Ni kitendo cha kutembea mara saba baina vilima viwili, Saffaa na Marwaa, kwa kuanzia Saffaa na kumalizia Marwaa.

Mafunzo:

- Inatufunza utii kwa Mwenyezi Mungu (s.w).
- Inatufunza ushujaa na uhodari katika kupigania Uislamu.
- Inatufunza kuwa tayari kuhangaika kwa ajili ya kupigania Uislam

vi. Kupiga Kambi Mina, Mwezi 8 Dhul-Hijah.

Ni kitongoji kilichopo baina ya Mji wa Makkah na bonde la Arafah. Mahujaji hulala hapo na kuondoka kesho yake kuelekea Arafa asubuhi.

Mafunzo:

- Huwafunza mahujaji kuwa askari waliotayari kuacha kila chaokwa ajili ya kupigania Uislamu.
- Huwafunza mahujaji kuwa askari wanatiifu kwa Muumba wao.

vii. Kusimama Arafah, Mwezi 9 Dhul-Hija.

Arafa ni bonde lililo baina ya vilima vitatu. Mahujaji husimama humo tangu Swala ya Adhuhuri hadi Magharibi na kuomba dua mbali mbali.

Mafunzo:

- Huwakumbusha mahujaji mkutano mkuu wa siku ya Kiama.
- Hutufunza kuwa hakuna aliyebora ila amchae Mungu zaidi.
- Kisimamo cha Arafa kinatufunza usawa baina ya wanaadamu.

viii. Kupiga Kambi Mzidalfah.

Mzidalfa ni kitongoji kilicho baina ya Mina na Arafa. Mahujaji huswali Magharib na Isha hapo na kuelekea Mina mara tu baada ya swala Alfajir

Mafunzo:

- Huwafunza mahujaj kuwa askari waliojiandaa kupambana na maadui wa Mwenyezi Mungu (s.w).
- Kuomba dua hutufunza kuwa ushindi ni kwa nusura ya Mungu tu.

ix. Kutupa Mawe Mina katika Minara Mitatu, Mwezi 10, Dhul-Hija.

Ni kitendo kinachofanyika kuanzia Mwezi 10-13, Dhul-Hija kwa kutupa vijiwe saba kila Mnara.

Mafunzo:

- Zoezi hili huwafunza mahujaj ujasiri wa kupambana na maadui zao.

- Huwafunza mahujaj pia mbinu na silaha katika kupigana na maadui.

x. Kuchinja mnyama Mwezi 10, Dhul-Hija.

Katika siku hii Hajj atanyo nywele, kuchinja mnyama, kutupa mawe kwenye Minara mitatu na kutufu Tawaful-Ifaadha (tawafu ya nguzo).

Mafunzo:

- Zoezi hili linawafunza mahujaj utiifu kwa Mwenyezi Mungu (s.w).
- Tunajifunza kuwa tayari kuua au kuuawa kwa ajili ya Mungu.
- Inatufunza kuwa na subira na kujitoa muhanga katika kunusuru din

xi. Siku za Tashriiq Mwezi 11-13, Dhul-Hija.

Ni kilele cha ibada ya Hija, ambapo mahujaj hupumzika kambini na kutathmini vitendo vyote walivyovifanya

Mafunzo:

- Huwafunza mahujaj kumkumbuka Mungu baada ya kazi ngumu.
- Huwafunza askari kuwa na mazingatio na yale waliojifunza kwayo

xii. Tawaf ya Kuaga (Tawaful-Widaa).

Ni Tawafu ya kuaga baada ya ibada ya Hija kukamilika na kujiandaa kurejea nyumbani kwa mahujaji.

Mafunzo:

- Huwafunza mahujaj kuwa viongozi (makhalifa) katika ardhi.
- Kuwa askari waliochukua mafunzo tayari kwa kuyatekeleza.
- Kuchukua ahadi ya utiifu kwa Mwenyezi Mungu (s.w).

Mgawanyo wa alama
Utangulizi alama (2)
Kiini alama (21)
Hitimisho alama (2)
J UMLA 25

(8)**Sababu za Uislamu Kupinga Dhana ya Kudhibiti Uzazi**

i. Sera ya kudhibiti uzazi inachochea zinaa ambayo ni njia chafu, mbaya na ina madhara makubwa katika jamii ya binaadamu.

ii. Sera ya kudhibiti uzazi ni kupinga na kuingilia mpango wa Allah (s.w) katika kupanga uzazi kwa njia ya ndoa.

iii. Dhana kuwa kuongezeka watu hupunguza riziki ni potofu na ni shirki, kwani ni kukana uwezo wa Allah (s.w) wa Kuruzuku na Kumiliki kila kitu.

iv. Kudhibiti uzazi ni kupoteza stara, hifadhi, huruma na mapenzi ya kweli baina ya mwanamke na mwanamume ambapo ni kinyume na utaratibu wa Allah (s.w).

v. Kudhibiti uzazi kwa hofu ya umaskini ni haramu, kwani kila kiumbe Allah (s.w) amekikadiria riziki yake.
vi. Sera juu ya kudhibiti uzazi inalengo la kuepuka jukumu la kuzaa, kulea na kuendeleza kizazi kwa utaratibu unaofaa.

vii. Sera ya kudhibiti uzazi ni kuchochea ngono huru, ukahaba, kuzaliwa watoto kiholela wasiokuwa na baba na kuondoa nafasi ya ndoa katika jamii.

viii. Kudhibiti uzazi ni dhana ya uchoyo, ubadhirifu na ulafi wa kutaka kumiliki na kutumia rasilimali, uchumi, siasa, n.k kwa watu wachache

Mgawanyo wa alama
Utangulizi alama (2)
Kiini alama (21)
Hitimisho alama (2)
J UMLA 25

(9)**Madhara ya Riba Katika J amii**

i. Riba inachafua nafsi ya tajiri, kwa kuwa bahili, mchoyo asiyetayari kuwasaidia maskini, wanyonge na wenye shida mpaka bila malipo yeyote ya ziada.

ii. Riba huchafua mali ya tajiri kutokana na njia haramu za kuchuma na kumiliki bila kuzingatia mipaka.

iii. Riba inakandamiza wanyonge, maskini na wenye dhiki na matatizo, wenye kuhitajia misaada ili waweze kuishi.

iv. Riba inaharibu uchumi na maendeleo ya jamii kwa kudhulumu kundi la maskini na kufaidisha kundi la matajiri.

v. Riba inahamisha uchumi, mali na rasilimali kutoka kwa maskini na wanyonge kwenda kwa matajiri.

Mgawanyo wa alama
Utangulizi alama (2)
Kiini alama (21)
Hitimisho alama (2)
J UMLA 25

SEHEMU E TAREIKH

(10)**Mbinu walizotumia Makafiri wa Kiqureish katika kuupinga Uislamu Makkah.**

- Maqureish wa Makkah walitumia kila mbinu kuhakikisha kuwa ujumbe wa Uislamu haufiki na kukubalika katika jamii yao, miongoni ni hizi zifuatazo;

a) Kumkatisha tamaa Mtume (s.a.w) na wafuasi wake.

- Makafiri walimdhihaki Mtume (s.a.w) na waumini wachache kwa

kumuita majina mabaya kama; mwendawazimu, mtunga mashiri,

aliyepagawa na mashetani, mchawi,

- Pamoja na kusimangwa huko Mtume (s.a.w) na waislamu hawakukata

tamaa, waliendelea kuutangaza Uislamu katika jamii yote.

b) Mbinu ya Vitisho na Kumkataza Mtume (s.a.w) kuulingania Uislamu kwa kumtumia Ami yake, Abu Talib.

- Wakuu wa Maquraish walimuagiza Abu Talib amkataze Muhammad

(mwanae) kuendelea kuutangaza Uislamu kwa kumbadilishia mtoto

mwingine la sivyo watamshambulia yeye na Muhammad (s.a.w).

- Mbinu hii haikufanikiwa badala yake Abu Talib alimuunga mkono

Mtume (s.a.w) kwa kazi yake baada ya kujua msimamo wake.

c) Mbinu ya Kumhonga na kumrubuni Mtume (s.a.w).

- Maquraish walitumia diplomasia kwa kumtaka Mtume (s.a.w) aache

kazi ya kuutangaza Uislamu kwa kumpa kati ya vitu vifuatavyo;

utajiri, mwanamke mzuri, madaraka au yote kama akitaka.

d) Mbinu ya kutaka kumlaghai Mtume (s.a.w).

- Maqureish walimletea Mtume (s.a.w) pendekezo kuwa washirikiane

katika ibada kwa awamu, mwaka mmoja wamuabudu Mungu mmoja

kwa pamoja na mwaka unaofuatia waabudu miungu yao pamoja pia.

- Hapo ndipo iliposhuka suratul-Kafiruun kuvunja pendekezo lao hilo la

kushiki katika ibada na Mtume akawasomea aya za Qur'an kukataa.

e) ***Mbinu ya Kuwatesa na kuwaua Waislamu.***

- Baada ya Maqureish kuona mbinu zote zimefeli, walianza kutumia

nguvu na mabavu kuwatesa, kuwanyonga na kuwaua Waislamu

waliong'ang'ani imani yao kama akina; Bilal bin Rabbah, Sumaiyya,

Ammar bin Yasir na baba yake, n.k

- Pamoja na mateso na mauaji, waislamu walibakia na msimamo wao

bila kutetereka na kukata tamaa.

f) ***Mbinu ya fitina, kuipinga Qur'an na kuzuia watu kukutana na Mtume (s.a.w).***

- Maquraish kwa kuhofia ujumbe wa Qur'an kuenea kwa watu,

walianza kuzuia watu kwenda kuonana na Mtume (s.a.w) kwa

kuwashawishi kuwa wasisikilize aya za Qur'an kwani ni uchawi,

mashairi ya Muhammad.

- Walizidi kufitinisha kwa kumzuia Mtume (s.a.w) kuswali na

kuwaambia watu kuwa Muhammad anapinga ibada na miungu wenu,

lakini mbinu na ushawishi wao ulifeli pia

g) ***Mbinu ya kuwatenga na kila huduma ya kijamii.***

- Maquraish walizidi kuongeza mateso kwa waislamu kwa kutenga na

kuwafukuza Makka na kwenda kuishi katika jangwa la Shi'ab

miaka

mitatu bila chakula wala mahusiano mengine ya kijamii.

- Mbinu pia ilifeli baada ya baadhi ya Maquiesh kuona huruma kwa

ndugu zao waliotengwa bila sababu za msingi, hivyo wakavunja

mkataba wa kuwatenga.

h) Mbinu ya kutaka kumuua Mtume (s.a.w).

- Maquish waliazimia kumuua Mtume (s.a.w) na rafiki yake Abubakar

(r.a) walipogundua kuwa waislamu wamehamia mji wa Madinah.

- Mbinu hii pia ilifeli baada ya Allah (s.w) kuwanusu katika pango yeye

na sahibu yake Abubakar (r.a).

Mgawanyo wa alama
Utangulizi alama (2)
Kiini alama (21)
Hitimisho alama (2)
J UMLA 25

(11) Dola ya Kiislamu ilianza kuporomoka nusu ya pili ya uongozi wa Khalifah, Uthman bin Affan (r.a).

- Wanafiki wakiongozwa na Abdullah bin Sabaa, walizua fitna na kumtuhumu kiongozi wa Dola na kuanzisha imani na itikadi potofu.

- Wapinzani wa Dola ya Kiislamu walijiimarisha na kutuma wajumbe watatu kutoka Misri, Basra na Kufa kwa Khalifa kuwasilisha malalamiko yao.

- Wajumbe hao walirejesha fitina kwao kuwa ujumbe wao umepuuzwa na Khalifa, hivyo wakaandaa jeshi la askari 1000 ili kuangusha utawala wa Uthman (r.a).

- Wakazi wa Madina walipambana na waasi hao lakini walizidiwa nguvu na hatimaye kuingia hadi Msikitini na kumuua Khalifa akiwa anasoma Qur'an.

Mgawanyo wa alama
Utangulizi alama (2)
Kiini alama (21)

Hitimisho alama (2)
JUMLA 25

(12) **Mafunzo yatokanayo na Mikataba ya 'Aqabah katika kuandaa ummah.**

i. Waislamu hawana budi kupanga mikakati na mambo yao kwa siri ili kutojulikana na maadui wa Uislamu na waislamu.

ii. Waislamu hawanabudi kutotangaza mikakati yao kwa maadui kabla ya kuitekeleza ili kuepukana na njama ya kukwamishwa.

iii. Waislamu hawanabudi kuwa na mipango na mikakati (long and short term strategies) ya muda mrefu na mfupi katika kuandaa Ummah.

iv. Uongozi bora na makini ni jambo la msingi sana la mwanzo katika kuusimamisha na kuuendeleza Uislamu katika jamii.

v. Waislamu wanaharakati wanapoazimia jambo lolote katika kuusimamisha na kuuendeleza Uislamu, hawana budi kujifunga nalo na kupeana kiapo cha utekelezaji.

vi. Waislamu hawana budi kutumia kila aina ya fursa itakayojitokeza katika kuusimamisha na kuuendeleza Uislamu katika jamii

Mgawanyo wa alama
Utangulizi alama (2)
Kiini alama (21)
Hitimisho alama (2)
JUMLA 25

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THE NATIONAL MUSLIM COUNCIL OF TANZANIA
ZAIM EDUCATION DEVELOPMENT LIMITED

FORM FOUR ZAIM SCHOOLS MOCK EXAMINATION

ENGLISH LANGUAGE
MARKING SCHEME
(For Both School and Private Candidates)

SECTION A (10 Marks)

COMPREHENSION AND SUMMARY

Answer **all** questions in this section.

1. (a) Choose the most correct answer and write its letter in your answer booklet.
5 marks, 0.5mark@

Answers for i- iv @0.5mark

i	ii	iii	iv	v	vi	vii	viii	ix	x
B	A	C	C	B	B	E	A	C	B

2. Read and summarize the following passage two sentences. **5marks 2.5 mark @ sentence**

Candidate must identify the topics of the passage before a candidate summarizes the passage into two sentences. The candidate must use as many words of his/her own as possible to mark understanding of the passage. The two sentences must be logical and linked to the ideas raised in the given passage. Candidate must retain the debate nature of the passage. A candidate becomes incorrect when a candidate provides answer to question instead of summary.

For instance the passage has two interrelated variables/issues of discussion. The passage begins to introduce dependence on foreign support in question form then finishes with the threats of such dependence. The two sentences can therefore be of this or any similar kind

It is unlikely for countries in our developing world to maintain their independence while subordinating our economies to rich countries and international business institutions. It is more unbelievable that poor countries can retain true sovereignty while caught up in the realm of foreign aids.

SECTION B (20 Marks)

PATTERNS AND VOCABULARY

Answer **all** questions in this section

3. **4marks-0.5mark@**

i. took ii. nationalize iii. embraced/embracing iv. created v. villagilization/ villagilisation vi. settled vii. collective viii. maximize

4. **4marks- 0.5mark@**

(a) Mate (b) display (c) Pangaea (d) ancestors (e) managed (f) beg (g) pursue (h) similar

5. In each of these sentences one of the words is wrong. Identify and correct them by writing the correct spelling for each word. **4marks- 0.5mark@**

Answers

a) The furniture are sold at cheaper price **[the noun Furniture does not take 's' in its plural form]**

b) The equipment are needed at workplace today **[the noun 'equipment is plural by nature hence needs a plural verb]**

c) We are watering the garden now. **[the word water is used as a verb hence 'ing' must be added to it to correspond with the time of reference expressed by an adverb 'now']**

d) I took a piece of advice from J uma last week **[the noun 'advice' does not agree with the adjective 'loaf' hence the word piece must substitute the word loaf]**

e) Girls have to liberate on their own from boys' traps **[the subject ' Girls' is plural hence the verb 'has' has to replaced by the plural verb ' have']**

f) Ali met me for a discussion last evening. **[the verb meet conflicts with the simple past time of reference proclaimed by the sentence hence the verb' met' must be used]**

g) Much information about a town is gathered through narrations. **[the word information does not take 's' in its form hence must drop the 's']**

h) Both house girls and house boys know sufferings [**The verb know must drop 's' form since the subject is a plural one]**

6. **4marks- 0.5mark@**

a) Vision b) purpose c) congregation d) legend e) delegates f) sex g) ballad

h) comrades

7. Re-write the following sentences according to the instructions given after each.
4marks-1mark@
   b) Halima is tall. Asha is tall
   c) Neither Juma nor Ali hate Ugali
   d) Musa promised that he would come the following /net day/ day after
   e) Our country is invited by the Democratic Republic of Congo to state celebration

SECTION C (30 Marks)

LANGUAGE USE

Answer questions **8, 9** and **two (2)** of the **four (4)** given alternatives in question **10**.

8. **5marks-1mark@**

Sentence number	1	2	3	4	5
Letter	**D**	**B**	**A**	**E**	**C**

9. Match the items in **List A** with responses in **List B** to make meaningful sentences by writing the letter of the correct response beside the item number in the answer booklet(s) provided.
5marks-1mark@

Number	i	Ii	Iii	Iv	V
Letter	**H**	**G**	**C**	**B**	**D**

10. Answer **two (2)** of the **four (4)** alternatives A, B, C and D.

A. Write a story tilted A DAY SHE WILL NEVER FORGET (At least 250 words)
10marks

Candidate must first start by writing the title of the story in capital letters and locate the title at the center. **A well written title deserves one (1) mark**. If a candidate fails to capitalize the title or misallocates the title the candidate should miss the mark. In short a candidate must copy the title as it is written in the question.

The introduction should not include definitions of terms but a candidate must create an interesting beginning of a story since this question is all about story writing hence story telling styles are used. **A relevant and correct organized introduction deserves two (2) marks**

Candidate should continue to give relevant background to the story creating imaginary setting and incidences to picture realities. The story should be told in third person reference which includes using a girl, a woman or a pronoun she. A

candidate should not use any other references apart from this third person point of view. **The logical continuity of the story deserves five (5) marks**

Candidate must end the story and at this juncture a candidate reveals the unforgettable experience a given girl or woman went through. The whole presentation should picture the past hence candidate must present the story in past tense. **A logical end gets two (2marks)**

Example

A DAY SHE WILL NEVER FORGET - the title 1mark

"Life is a classroom for more learning and human are pupils learning each day till they die".

Amina used to hear her father telling her elder brothers and sisters when Amina was still young. At that time such statement had little or no meaning in ears and minds of Amina as she thoughts that life might have more than what her parents said. Amina thought parents were only threatening children for no practical reasons.
[End of a story introduction 2marks]

With this attitude in mind Amina grew as a solid and expectant girl. She lived a kind of life full of expectations just like other young girls in many societies. Amina continued to work hard in trade as her family was well known for trade for ages in her Mtakuja village. Trade gave her fame and wealth such that she became more proud of herself and her efforts. **[Continuity of the story begins]**

One evening Amina received request to supply clothes she had bought from town to some of her clients at a nearby village. Her parents advised her not to go alone at that hour. Her brothers were not there so the parents insisted that Amina waits for her brothers who were expected to return from town the following day. **[Continuity of the story]**

As a business girl Amina calculated the loss she would incur if she loss the tender. She was not able to tolerate so she resented her parents' advice. Amina went alone. **[Continuity of the story ends- 5marks followed by suspense]**

The next day Amina found herself waking up at a hospital ward. She was attacked by gangs, beaten and raped by gangs on her way back home. Amina cried a lot on bed as she lost all her money in one life incidence and she feared she might have contacted AIDS. **[Ending of the story 2marks]**

B. Imagine you are the new appointed Regional Commissioner of Kigoma region. In not less than 200 words write a first speech you would deliver in an official welcoming ceremony at aqua lodge beaches. Focus on your plans as a new president Magufuli's appointee.**[10marks]**

Candidate should write an introduction for salutation and acknowledgement of official formalities. The salutation should form a block.
In such introduction hierarchies of authorities should be logical. Logistics of regional commissioner visit must be seen such as the presence of the district commissioner, Regional Police Commander, town Director, councilors, and etcetera.

2marks

Then a candidate should write an introduction to speech which is an opening word. Formally in this introduction section of speech candidate should express the pleasure for visitation not thanks giving. **1.5marks**

The body of the speech has to include plans the Regional Commissioner has for the region. A candidate must not write the plans in hint format but as part and parcel of a talk. **5marks**

The candidate must end the speech with word of thanks and any other relevant comments like pledging for cooperation. **1.5marks**

Example

Honorable District Commissioner, Kigoma district,
Honorable Town director, Mr Lukiri,
Honorable Regional Police Commander, RPC. **[This is salutation: LOGISTICS observed 2%]**
Members of Parliaments,
The councilors,
Dear Citizens of Kigoma,
Ladies and Gentlemen,
Protocol observed/ Good evening.

[Below starts an introduction to a speech, that is an opening word to the audience]

I'm so pleased and honored to stand before you this day as new regional Commissioner of your region. Brothers and sisters, I was appointed by his excellence the head of state honorable Doctor John Pombe Joseph Magufuli on the 2nd day of August 2019. My appointment was effected two days later.

My names are Ally Musa Hassan Kibakuli. Formerly I was a chief Advisor of regional commissioner of Lindi, a region whose challenges resemble much these of Kigoma. I have been in administration for more than ten years now and throughout this period I have gathered much experiences and I can say I'm in a mood and position to serve you my people.

I know it is not easy to work for people and with people. There are always challenges on the way to limit achievements. I know there will always be some problems somewhere that is why I must seize this platform to channel my priorities that will define my term of service and judge my administration.

[The introduction and opening word ends here 1.5marks]

[The main body begins here below- the plans are well spoken - 5marks]

Dear citizens and all listening and watching us via various media, my focus which I'm sure will be the focus of us all as a region will be on creating more jobs to absorb more people particularly youth in labour market. I want to solve unemployment which is hitting our region. We will achieve this by encouraging

entrepreneurship and we shall create 200 new jobs to top to the present 150 jobs available. This will help us employ the 60 per cent of the unemployed population.

We are going to secure more funds from donors by encouraging Youth and Women organizations to seek for fund from donors. We are going to promote cooperatives such as SACCOS. We are going to strictly monitor financial resources to eliminate embezzlement. We are going to repair damaged roads and build new ones.

We shall also pay much attention on social services provision. The region will ensure the fund allocated for health and education centres is not misused. We shall also allow citizens in puts to better the provision such services.
[The main body ends]

[The end of the speech begins here below 1.5marks]
These and many other future plans need support. I therefore ask for support from my subordinates and every region dweller. Together we will make Kigoma a better place to live and a better home for the coming generation.

Thank you very much for your time. Thanks a lot for listening and watching. Thanks

I wish you all the best in working hard to build our region Kigoma and our lovely nation Tanzania.

C. Write an invitation card to request your sisters' friends to attend on your niece's birthday dinner party. Also write telephone messages from those who will call for excuses. Use fictitious names. **[10marks]**

A candidate should design a card which must have eight important elements and each element weights 1**mark** to make a **total of 8marks**. **Two (2) other remaining marks shall be awarded to candidate if a candidate manages to make appropriate** card design that is to say, a card must look as it supposes to look like in the field of card design.

The eight elements are as follow;

- Name of host
- Invitational (requests the pleasure of your company, cordially invites you to)
- Event type/name/purpose (e.g. dinner party)
- Date
- Hour **1mark@**
- Place
- Reply instructions (e.g., reply card or phone number)
- Direction

D. Write a letter to the Editor of Guardian Newspaper on the challenges of

democracy in Tanzania. Use fictitious name and address. **[10marks]**

A candidate must include in his/her letter the following eight elements and the distribution of marks shall be as shown in the example given bellow this outline of elements.

1. Sender's address: The address and contact details of the sender are written here. Include email and phone number, and etcetera.
2. Date: The date is written below the sender's address after Leaving one space or line.
3. Receiving Editor's address: The address of the recipient of the mail i.e. the editor is written here.
4. Subject of the letter: The main purpose of the letter forms the subject. It must be written in one line. It must convey the matter for which the letter is written.
5. Salutation (Sir / Respected sir / Madam)
6. Body: The matter of the letter is written here. It is divided into 3 paragraphs as follows -
Paragraph 1: Introduce yourself and the purpose of writing the letter in brief.
Paragraph 2: Give a detail of the matter.
Paragraph 3: Conclude by mentioning what you expect from the editor. (For example, you may want him to highlight the issue in his newspaper / magazine).
7. Complimentary Closing
8. Sender's name, signature and designation (if any)

A Working example

33, J al Vihar
Wazirabad, Nazareth Kigoma – 33 **[0.5mark]**
Dated: 17 J uly 2019 **[0.5mark]**
The Editor
The Guardian Newspaper **[0.5mark]**
Dar es Salaam.
Subject: Need to get rid of the challenges of democracy in Tanzania **[0.5 mark]**

Dear Editor **[0.5 mark]**

I am Radha G, member of New Light Tanzania. I am writing to you in order to highlight the deteriorating condition of democracy in Tanzania. **[0.5 mark]**

Democracy in a country Tanzania is getting weakened with the change of times. The citizens are blaming for such deterioration of democracy. The lack of freedom to expression is hitting every man and women to the point that some religious leaders have dared to request the head of state to lessen the use of force. **[2marks]**

The people have been demanding a respect for the constitution of the United Republic of Tanzania. The state organs have not yet responded to the repeated requests and

cries of the people. **[2marks]**

I request you to highlight the problem in your newspaper and arouse public interest. We all need to get together in order to retain the rule of law and true democracy in our country. **[2marks]**

Thank You
Yours sincerely **[0.5 mark]**

Radha G
Member of New Light Tanzania **[0.5mark]**

11. How does this poem describe the position and roles of women in our societies? (Give eight points**). [20marks]**

Candidate must introduce the question by giving relevant back ground information on women lives as it is evident in many societies. The introduction must be a paragraph of its own right. That is must be presented as a separate paragraph. **This beginning deserves 1mark**

Then candidate has to introduce the written poem by stating its general/ main focus. A candidate must do this in a second paragraph to avoid over generalization. **This second paragraph weights only 1mark**

The main body of this question must be composed of eight complete paragraphs. Each paragraph must deal with only one role or position the written poem has raised. Each well explained role/position deserves **2marks. The whole main body weights 16marks**

Candidate should end the presentation with a conclusion which shall speak all that the poem s ays about women in one language. A relevant conclusion gets **2marks**

Some of the points include 2marks@

a) Women are treated as source of leisure.
   - **A candidate must explain and then give pieces of evidence from the given poem to justify. If a candidate does only one part of the two tasks 1or 0.5mark out of two(2) marks should be given**

b) Women are reduced to child bearing role only
   - **A candidate must explain and then give pieces of evidence from the given poem to justify. If a candidate does only one part the of two tasks 1or 0.5mark out of two (2) marks should be given**

c) Women lack right to own properties.
   - **A candidate must explain and then give pieces of evidence from the given poem to justify. If a candidate does only one part the of two tasks 1or 0.5mark out of two (2) marks should be given**

d) Women are treated as mere properties, Sold and bought via bride price
   - **A candidate must explain and then give pieces of evidence from the given**

poem to justify. If a candidate does only one part the of two tasks 1or 0.5mark out of two (2) marks should be given

e) Women are made servants to men, handled like tools, work as slaves, sold slaves
   - **A candidate must explain and then give pieces of evidence from the given poem to justify. If a candidate does only one part the of two tasks 1or 0.5mark out of two (2) marks should be given**

f) Women are oppressed, suffer from continued oppression.
   - **A candidate must explain and then give pieces of evidence from the given poem to justify. If a candidate does only one part the two tasks 1or 0.5mark out of two(2) marks should be given**

g) Women are neglected, marginalized in the society. They are deprived of their rights
   - **A candidate must explain and then give pieces of evidence from the given poem to justify. If a candidate does only one part the of two tasks 1or 0.5mark out of two(2) marks should be given**

h) Women are discriminated
   - **A candidate must explain and then give pieces of evidence from the given poem to justify. If a candidate does only one part the of two tasks 1or 0.5mark out of two(2) marks should be given**

i) Women are despised., regarded as inferior sex for ages
   - **A candidate must explain and then give pieces of evidence from the given poem to justify. If a candidate does only one part the of two tasks 1or 0.5mark out of two(2) marks should be given**

Conclusion

The main theme of a conclusion should revolve around this idea "Women know sufferings for generations now". **2marks**

12. Poets compose poems not just for pleasure and entertainment but for addressing human social, political and economic issues. Using Song of Lawino and Ocol present eight points to verify the statement **[20marks]**

    Candidate must define the term POEMS not POEM. Candidate must say who POETS are. The candidate has to conceptualize poems in a complete paragraph since a paragraph must always address not more than one issue of discussion.

    The second paragraph has to briefly explain the poem Song of Lawino and Ocol in terms of the writer and the main focus of the poem or any relevant introductory comment about the book. Candidate shall also use this paragraph to introduce the question.

    The main Body is going to be of eight paragraphs. Each paragraph must address only one issue. **A relevant paragraph gets 2marks and therefore the whole main body shall weight 16marks**

Some of the issues the Book raises includes; 2marks@

a. The influence of Christianly and the mushrooming churches ;the poem cites

the Catholics, the protestants and the hypocrisy there in

b. The role of Formal education in alienating individual from tradition life as it is for Ocol
c. Illiteracy among Africans i.e Lawino
d. Political enemity. Ocol and his brother who are enemies just because of belonging to different political parties, CCM and CUF particularly in Zanzibar
e. Western cultural imperialism i.e Clementine who fights to look like a white woman.
f. Moral decay. Unethical dressing, public kissing, the sugar daddies and sugar mommies , the miniskirts in Tanzania etcetera
g. Generation gap. The antagonism between the new and the old generation
h. The class of African elite and their struggle for power and domination.

Conclusion

Relevant conclusion should spell one main striking issue that the poem talks about that is more evident to our societies. 2marks

13. 'Rural life in Africa is itself a separate world, a world with its own life styles and manner'. Using two plays of your choice provide four pieces of evidence from each play to substantiate the said analogy. **[20marks]**

In the first paragraph of an introduction candidate should say something about rural life. This has to take one paragraph. **[1mark]**

In the second paragraph candidate needs to introduce the needs of the question plus the two titles of the play candidate is going to refer. The candidate must identify the respective playwrights and the titles must be written appropriately. **[1mark]**

The main body must present eight features of village lives as portrayed by different plays. **2marks@**

Examples

a. In village there are few elites. Illiteracy is high. People know less about the external world
b. Villages are dominated by witchcraft beliefs
c. Villages lack most of modern technologies like camera, motor cycles
d. In villages Polygamy is an order of the day. This create extended families with poor life
e. There are few and inadequate social services in rural area. There few schools and in many cases no secondary school.
f. Most villages live communal life. People live as one community. Traditions are so powerful in rural areas.
g. In villages leaders abuse their power and there is no way to check power abuse. The police stations are absent in many places in villages. The village heads become the court and the judge.
h. In villages women are at more disadvantaged position than town.

Conclusion

The conclusion should focus on one general feature of village life with reference to the two plays. [2marks]

14. 'In human society misunderstanding is inevitable and this emanates from the roots of the problem which are evident in many societies'. With reference to two novels discuss the eight major roots of misunderstanding, four roots from each novel. **[20marks]**

In introduction part candidate must explain what misunderstanding is. The candidate has to show the manifestation of misunderstanding i.e. political, social, ideological etcetera. This has to be done in one paragraph

The second paragraph of an introduction has to practically introduce the question 'roots of misunderstanding' Candidate should show the titles of the novel and novelists he/she is going to refer to. **[A well organized introduction gets 2marks]**

There must be eight paragraphs in the main body. Each paragraph handles one root of misunderstanding and if well explained **two (2) marks are awarded to each paragraph**. The whole main body weight **16marks**

Some of the roots of misunderstanding may be

a. Betrayal
b. Traditionalism
c. Modernization **2marks @**
d. Formal Education
e. Cultural imperialism
f. Tribalism g. Conservatism h. Stratification i. Marginalization, discrimination J. Oppression, humiliation and exploitation

Conclusion

Candidate must give one or two key roots which are the main causes of misunderstanding [2marks]

BISMILLAH RAHMAN RAHIIM

THE NATIONAL MUSLIM COUNCIL OF TANZANIA
ZAIM EDUCATION DEVELOPMENT LIMITED

FORM FOUR ZAIM SCHOOLS MOCK EXAMINATION

012 **HISTORY**
MARKING SCHEME
(For Both School and Private Candidates)

SECTION A (20 Mark

Answer **all** questions in this section.

1. **This question number one weights 10 marks 1mark@**

i	ii	iii	iv	v	vi	vii	viii	ix	x
D	B	C	E	A	E	A	E	B	A

2. **This question weights ten (10) marks 1mark@**

i	ii	iii	iv	v	vi	vii	viii	ix	x
H	B	C	D	E	F	G	I	J	No match. There was an omission error. A candidate who discovers that the question has no answer deserves 1mark. Candidate who will write any answer deserve no mark.

SECTION B (20 Marks)

Answer **all** questions in this section.

3. (a) Draw a sketch map of Africa and locate **7.5marks, 1.5marks@**
   i. The trans Atlantic caravan routes
      Using a pencil candidate must show the routes on the map connecting the three continents; Africa, America and Europe during triangular slave trade **[Trans Atlantic trade implies Triangular trade or Trans Atlantic slave trade] 1.5marks**

   ii. The cape colony
      Candidate must locate the Cape of Good Hope **1.5marks**
      This is at the horn of South Africa, a land at the edge of South Africa meeting the Atlantic Ocean where whites settled on their way to Asia.

iii. The land of Northern Rhodesia
Candidate must locate Northern Rhodesia **1.5marks**
This is Zambia hence a student must locate Northern Rhodesia- Zambia appropriately

iv. The city state and country where Fort Jesus was built by the Portuguese
Candidate must locate the city state- Mombasa city **1.5marks**
This is Mombasa there in Kenya, a coastal region of Kenya.

v. The two African countries which were not colonized.
Candidate must locate the two countries on the same map- Liberia and Ethiopia **1.5marks**
Liberia is in West Africa. Ethiopia is in North East Africa bordering Eritrea.

(b) Mention five coastal towns in East Africa which became famous because of the commercial contacts between the people of East Africa, Middle and Far East **2.5marks, 0.5@**
Candidate must list relevant coastal towns of East Africa such as **Kilwa, Pate, Lamu, Mogadishu, Sofala. The towns must be those that rose at the time Kilwa was the great coastal city state.**

4. (a) Arrange the following statements in chronological order by writing number 1 to 5 beside the item number.

i	ii	iii	iv	v
5	3	2	4	5

5marks, 1mark@

(b) The items (i- iv) consist of four sentences which form a complete historical meaning. One of the sentences (A- E) is **missing.** Identify the missing sentence and write its letter beside the item number.

i	ii	iii	iv	v
C	D	C	D	B

5marks, 1mark@

SECTION C

Answer **three (3)** questions from this section.

5. Discuss the cause of the Boer trek and its consequences to the people of South Africa.
Give three points each side
In an introduction candidate must define BOER TREK. Candidate has to periodize the trek. The introduction weights **03 Marks**

The main body must have six points. There should be three causes at the beginning of the main body followed by the three negative impacts of Boer trek [consequences].

Some of causes for Boer trek

a. Land shortage at the cape **2.5marks @**
b. British domination at the cape
c. The deteriorating relationship between the Boers and the British at the cape

Some of the consequences of the Boer trek

a. Outbreak of wars between Boers and the native Africans
b. Enslavement of natives by the Boers
c. The Anglo Boer wars after discovery of minerals by the Boers

There shall be six paragraphs in the main body each weighting **2.5 marks** making a total **of 15marks for the whole main body.**

The conclusion should focus on either one main cause or one main consequence of the Trek [2marks]

6. Account for the rise and fall of Mali Empire in Western Sudan. Give three points each side

In introduction candidate must define Mali kingdom, locate the empire and periodize Kingdom's the emergence. The empire founders need to be mentioned.

The introduction weighs **3 marks and should define Mali Empire, identifying founders of the empire and time for the rise of the empire**

The main Body shall contain the three cause/ factors for rise and three factors for the collapse of the Mali empire. There shall be a total of six paragraphs in the main body alone each paragraph deserves **2.5 marks.** The whole section of the main body gets **15 marks**

Some factors for the rise of Mali Empire

a. Weakening of Ghana empire
b. Role of Sundiata of Keita clan who unified the clans
c. The Mali Empire control of Gold trade

i. 2.5marks@

Some factors for the collapse of Mali Empire

a. The death Mansa Musa i.e there were no clear laws to decide who would succeed him.
b. The rise of Songhai
c. Loss of control of trade **2.5marks@**

The conclusion should focus on either one main cause for rise or fall of the empire **[2marks]**

7. Why did it take too long to end slave trade in East Africa? Give six reasons

Candidate introduction has to involve definition of slave trade in the first paragraph of the introduction.
The second paragraph of introduction must say something about slave trade abolition and slave trade abolition in East Africa. Candidate should periodize slave trade abolition in East Africa or Africa in general. **The complete introduction gets 03 marks**

The Main body has to have six paragraphs each paragraph digesting one factor that made delay in slave trade abolition in East Africa.

Some of the factors for delay

a. Abolitionists were confined on the coast, the interior continued to trade **2.5marks@**
b. Britain had no enough patrol ships to effect the treaties with the sultanate
c. Clove and coconut plantation in Zanzibar continued to demand slave labour
d. French took slaves to work in plantations and forced them to sign as free labour
e. The external Arabs world continued to take slaves as domestic servants.
f. Local chiefs benefited from trade were reluctant to leave the trade

In the conclusion candidate must suggest with evidence one/two main factor(s) which contributed much to such delay in abolition of slave trade in East Africa- **[2marks]**

8. With examples elaborate the six weaknesses of colonial education.
In the introduction candidate must explain clearly COLONIAL EDUCATION in a complete paragraph [3marks]

The main body must present six shot falls of colonial education in six different paragraphs.

Some of the weaknesses of Colonial education include;

Racism

a. Pyramidal i.e. has many school drop outs
b. Irrelevance i.e syllabus were set in Europe and for Europe **[2.5marks]**
c. Theoretical i.e could not give practical skills
d. Un even distributed
e. Bookish i.e could not promote production

The conclusion should say one main limitation of the colonial education [2marks]

9. Decolonization in Portuguese colonies in Africa was tough and complicated. The colonies under Portuguese opted armed struggle instead of peaceful means.

Why such colonies waged armed struggle? Give six points

In introduction **DECOLONIZATION** must be defined and forms of decolonization stated. Candidate has to periodize decolonization in Africa
In the second paragraph candidate must give general information about Portuguese colonies with some examples of such Portuguese colonies [Angola, Mozambique, and etcetera] being mentioned. Candidate has to introduce the needs of the question. The complete introduction gets 3marks

The Main Body must present with concrete examples reasons for armed struggles in Portuguese colonies. The maximum reasons are six hence **each well grounded reasoning gets 2.5marks to make a total of 15marks for the main body alone**

Some of the points include

a. The banning of political parties and activities
b. Ignorance of the Portuguese on the possibility to rule the colonies by neocolonialism Killing of freedom fighters such as Edward Mondlane
c. Harshness of the Portuguese **2.5marks @**
d. The poor nature of Portuguese economy i.e c
e. Richness of colonies e.g Angola oil deposits
f. Portuguese took the colonies as over sea provinces [extra point]

Conclusion

Candidate must present one main general reason for armed struggle in Portuguese colonies **[2marks]**

10. Explain six problems facing post colonial African societies whose roots trace back to colonial era.

In the introduction candidate has to explain the concept **POST COLONIAL AFRICAN SOCIETIES**. Candidate should periodize independence of many African countries i.e. the 1960's as the year of independence. The complete and relevant introduction gets 3marks
The main body has to cover six problems in post colonial Africa in six different paragraphs. Each well composed paragraph gets 2.5marks making a total of 15marks for the main body alone.

Some of the problems include;

a. Shortage of social services i.e health and educational services
b. High illiteracy
c. Ineffective physical infrastructures i.e roads
d. Disunity due to tribalism and ethnicity i.e ethnicity intensified by colonialists Boundary conflicts i.e boundaries fixed by colonialists **2.5marks@**

e. Malnutrition diseases, famine and hunger i.e food shortage due to emphasis on cash crop
f. Economic crises i.e capitalist crises

The conclusion has to add another problem to close the Essay.[2marks]

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THE NATIONAL MUSLIM COUNCIL OF TANZANIA
ZAIM EDUCATION DEVELOPMENT LIMITED

FORM FOUR ZAIM SCHOOLS MOCK EXAMINATION

024 LITERATURE IN ENGLISH
(For Both School and Private Candidates)

MARKING SCHEME

Time: 2:30Hours Friday, 28th June 2019 p.m.

SECTION A (40 Marks)

THEORIES OF LITERATURE

Answer two (2) questions from this section.

1. Discuss the four advantages and four disadvantages of written literature over oral literature **[20marks]**

   Candidate must define oral and written literature in a separate paragraph in his or her introduction. Relevant definitions and well composed introductory paragraph gets (2marks)

   In the main body candidate should give merits and demerits of written literature. The main body is going to be composed of eight separate and complete paragraphs. The first four paragraphs of the main body shall digest the merits of written literature and the four remaining paragraphs shall address the demerits of written literature. **Each well composed paragraph gets 2marks to make a total of 16 marks for the main body alone.**

   SOME OF THE MERITS OF WRITTEN LITERATURE INCLUDE THE FOLLOWING

(a) Free from biasness ,i.e more objective one
(b) Develops writing *2MARKS @*
(c) Stores literary work for long without distortion
(d) Is more free from errors since writer makes enough preparation

SOME OF THE DISADVANTAGES ARE

(a) Expensive
(b) Discriminates illiterate audience
(c) Does not allow immediate modification *2MARKS@*
(d) Is not as much alive as oral literature

The conclusion must talk of the general merit or demerit of written literature [2marks]

2. (A) Write short notes on the following literary concepts **[10marks]**
In writing short notes candidate should write more than one sentence on the concept as it is seen below. A candidate is incorrect if he/she just gives one sentence hence marker should award half of the marks allocated to such candidate who fails to write NOTES.

i. Elegy
Elegy is a poem prepared for mourning the death of an individual. An Elegy poem is composed for mourners. **[2marks].There should be more than one sentence to compose notes unless award half of the two marks**

ii. Sonnet
Sonnet refers to a poem of 14 lines written in iambic pentameter. In this type of poem each line has 10 syllables. A Sonnet is mainly a one stanza poem **[2marks].There should be more than one sentence to compose notes unless award half of the two marks**

iii. Lyric poem
Lyric poem is a poem that gives strong feelings. It is a formal type of poetry which expresses personal emotions or feelings, typically spoken in the first person **[2marks].There should be more than one sentence to compose notes unless award half of the two marks**

iv. Tragedy drama
Tragedy drama is a drama in which the hero ends up into suffering. It is a type of drama with hard and terrifying experiences for the central figure of a story **[2 marks].There should be more than one sentence to compose notes unless award half of the two marks**

v. Literary criticism
Literary criticism is the art or practice of judging and commenting on the qualities and character of literary works. Literary criticism denotes a process of

evaluating, assessing effectiveness of literary works **[2marks].There should be more than one sentence to compose notes unless award half of the two marks**

(B) Briefly explain each of the following concepts as applied in literature **[10marks]**

In writing Brief explanations candidate should write more than one sentence on the concept as it is seen below. A candidate is incorrect if he/she just gives one sentence hence marker should award half of the marks allocated to such candidate who fails to Explain in brief.

i. Resolution

Resolution is the part of the story's plot where the main problem is resolved or worked out. The **resolution** occurs after the falling action and is typically where the story ends.**[2marks].There should be more than one sentence to compose brief explanations unless award half of the two marks**

ii. Climax

Climax is the highest or most intense point in the development or resolution of something; culmination. It is a decisive moment that is of maximum intensity or is a major turning point in a plot. **[2marks].There should be more than one sentence to compose brief explanations unless award half of the two marks**

iii. Crisis

Crisis means decision or dilemma. In a **story**, it's the do-or-die moment, that last chance where the protagonist must gather all their strength, ingenuity, resourcefulness, and courage in one final effort to defeat the opposing forces guarding the prize. **[2marks].There should be more than one sentence to compose brief explanations unless award half of the two marks**

iv. Prologue

Prologue is an opening to a story that establishes the context and gives background details, often some earlier story that tie into the main one, and other miscellaneous information. It is the opposite of epilogue **[2marks].There should be more than one sentence to compose brief explanations unless award half of the two marks**

v. Mystery

Mystery is defined as something that is a secret, something where there is no clear explanation, something difficult to understand or explain or something unexplainable or unsolvable. **[2marks].There should be more than one sentence to compose brief explanations unless award half of the two marks**

3. What is a play? How are plays unique? Give eight areas demonstrating uniqueness of play and drama
   Candidate has to define drama as part of an introduction. This should be done in a paragraph. A paragraph should be a unit of complete sentences and not otherwise and a sentence must be demarcated by full stop. There is no citation of titles here. The complete introduction carries two (2) marks

   There after follows a main body of eight paragraphs each paragraph treating one point in detail. Here below are some of the points candidates are expected to use to demonstrate the uniqueness of drama. The candidate should focus on the form of the genre not the content. **Each point gets 2marks marking a total of 16 marks for the main body alone.**

   (a) Cast of characters
   (b) Monologues
   (c) Dialogues
   (d) Actions
   (e) Acts and scenes ***2MARKS @***
   (f) Playwrights
   (g) Stage direction
   (h) Use of shortest sentence e.g one word sentence: YES! No!
   (i) Use of paralinguistic features i.e body movements

   Conclusion
   Candidate must give one general point which proves the uniqueness of play as a literary genre- 2marks

SECTION B (20 Marks)

PLAYS

Answer **one (1)** question from this section.

4. Relate the stories and incidences from the plays you have read with the realities people witness in daily life. Give four points from each play.

   Candidate must define plays in his/her introduction. Candidate should comment on plays and realities plus naming the titles the candidate is going to use. A complete introduction gets 2marks

   The candidate has to present eight incidences read from the plays of choice that can be seen in real life. Each incidence must be treated in a separate paragraph. A complete and well explained paragraph deserves 2marks making a total of 16 marks for the whole main body alone.

Example of incidences

a) Presence of only primary school (s) in Illujinle village i.e. . no secondary schools
b) The emphasis on Virginity of girls
c) The practice of polygamy in rural areas **2MARKS@**
d) Village girls wondering as they see camera and motorcycle
e) Young villagers leaving village in swamp dwellers
f) Religious leaders living luxury lives at the midst of poor believers
g) Villages dominated by the old people
h) Coming of beggars

Conclusion
Candidate must give general point which proves the relevance of the plays to real life 2marks

5. With references to two plays you have read, show how the conflicts found in the readings have impacted individuals. Give four points from each play.

   In introduction candidate has to define the term conflicts and classify conflicts. Candidate has to name the titles he/she is going to use. **A complete introduction gets 2marks**

   The main body should have eight impacts of conflict four impacts from each play. Each well treated effect in a paragraph gets two (2) marks making a total of 16 marks for the whole main body alone.

   Examples of points
   a) Loss of life
   b) Disunity
   c) Betrayal ***2MARKS@***
   d) Builds hatred
   e) Builds spirit of revenge
   f) Family separation
   g) Accelerates injustices
   h) Leads to violation of human rights

   Conclusion
   Candidate must give general point explaining the effect of conflict the two plays reveal in their respective stories 2marks

1. Verify relevance of stage direction in effecting art and content in two plays you have read. Give four points from each play to support your answer.

Candidate has to define stage direction. Candidate should locate the where of stage direction in a play. Candidate has to list the titles he/she is going to use and their respective playwrights. A complete introduction gets 2marks

The main body should focus on the role of stage direction as part of Form in delivery of

themes and messages. Candidate must cite concrete examples from chosen plays There shall be eight points in eight paragraphs. **Each well presented point deserves 2marks hence the main body weights 16 marks. Each play must give four points in four paragraphs only.**

Some of the hints on the importance of stage direction that students will use to explain how stage direction reinforces themes and messages are here below;

a. Introduces new character
b. Show the exit of character e.g death
c. Describe the activities to be done by characters **2marks@**
d. Describe the appearance of the character
e. Tells the time of events e.g moring, noon
f. Show the place of event eg. In a room or open space
g. Guides body movement to give body language e.g (laughing)
h. Tells the mood of a character
i. Shows mimicry, monologues

Conclusion
Candidate must give general point explaining the significances of stage direction the two plays reveal in their respective stories- 2marks

SECTION C (20 Marks)
NOVELS AND SHORT STORIES

Answer **one (1)** question from this section.

2. Characters in literary works inspire and disappoint readers. How are your inspired by any two characters from two readings you have done? Give four points for each character

Candidate introduction must include definition of the term **CHARACTERS**. Candidate must go further to classify characters.

Candidate should introduce the needs of the question. He/she has to name the titles a candidate is going to refer. Candidate must identify two inspiring characters. **A complete introduction gets 2marks**

The main body must focus on the nature of character personalities and the way such personality motivate readers. Candidate must present the points in third person point

of view. **Each well presented personality deserves 2marks hence the main body weights 16 marks.**

Some of the points include the following

a. is so proud of her/his culture
b. Works very hard
c. Loves his/her people
j. Is ready to learn , has an open mind **2marks@**
d. Never take things for granted. Thinks and question things
e. Is well determined and committed
f. Respects women and fights for the underprivileged
g. Fights for change

Conclusion
Candidate must give one general point explaining the inspirations the two characters reveal through the respective novels- 2marks

3. Choose two readings from those read under this section and write down thematic similarities and differences. Give four points from each reading.

Candidate should not define anything here since there is no key term to define. Candidate is expected to give relevant comments which must revolve around themes. Candidate should show the novels he/she has chose to use. A complete introduction gets 2marks

In the main body candidate should begin with the similarities of themes (four similarities) in the chosen novels followed by the four differences in themes; there will be a total of eight paragraphs each weighting 2marks. **Each well presented point deserves 2marks hence the main body weights 16 marks**

The conclusion may either talk of the general similarity or difference of themes of the two novels chosen- 2marks

4. Discuss eight roots of conflict displayed in two readings of your choice by giving four points from each reading.

Candidate must define CONFLICT. Candidate has to classify conflict and name the

titles candidate has chosen to use. A complete introduction gets 2marks

The main body has to digest eight roots of conflict in eight different paragraphs. Each well discussed point of a paragraph gets 2marks hence the main body takes a total of 16marks

Some roots of conflict include

a) Betrayal
b) Exploitation
c) Oppression and humiliation **2marks@**
d) Violation of human rights
e) Economic crises like unemployment crises
f) Dispossession of means of wealth such as land
g) Sex abuse
h) Lust for wealth, power and fame

Conclusion

Candidate must give one main root of conflict that prevails in all the two references and societies. 2marks

SECTION D (20 Marks)

POETRY

Answer **one (1)** question from this section.

TEXT: Selected poems —Tanzania Institute of Education

Using two poems demonstrate the eight unique features of poetry as an independent genre of literature

Candidate must define poetry in introduction. Candidate must identify names of two poems candidate has chosen. A complete introduction gets 2marks

The main body should have eight characteristics of poetry drawing evidence from chosen poems. Each well explained characteristic deserves 2marks hence the whole main body shall take a total of 16 marks

SOME OF THE COMMON UNIQUE FEATURES OF POETRY ARE AS FOLLOW;

a) Rich in figures of speech
b) Written by poet/ poetess
c) Written in verse form

d) Has frozen language
e) Exercises poetic license ***2MARKS@***
f) Has economy of word
g) Has emotional charged language
h) Is the oldest literature as it began as songs
i) Organized in stanzas

Conclusion
Candidate should say in one point the overall uniqueness of poetry - 2marks

5. Show how the poet has succeeded to deploy figures of speech to reflect realities in two poems you have studied under this section. Give six points

Candidate has to define FIGURES OF SPEECH in the introduction. Candidate has to name the titles of the two poems candidate has chosen to use. Complete introduction takes 3marks

The main body has to involve analysis of six figures of speech in six different paragraphs showing how the figures of speech portray themes and messages. Each well explained figure of speech gets 2.5marks hence the main body weighs a total of 15marks. The figures of speech include similes, metaphor, irony, personification, satire, hyperbole, and etcetera.

Conclusion
Candidate must close the essay with one comment relevant to the main body presented- 2marks

6. Discuss the eight messages from any four poems you have read. Give two messages from each poem

Candidate should define the term MESSAGES. Candidate should identify names of four titles of poems candidate wishes to use. A complete introduction gets 2marks

The main body shall include eight paragraphs each paragraph with well stipulated lesson from four chosen poem. Each poem should give two messages. Every well explained lesson gets two (2) marks hence the whole main body shall weigh 16marks

Conclusion
Candidate must close main body discussion with general message from the four poems - 2marks

BISMILLAH RAHMAN RAHIIM
THE NATIONAL MUSLIM COUNCIL OF TANZANIA
ZAIM EDUCATION DEVELOPMENT LIMITED

FORM FOUR ZAIM SCHOOLS MOCK EXAMINATION

BIOLOGY THEORY
MARKING SCHEME
(For Both School and Private Candidates)

SECTION A(20 MARKS)

1. Multiple choice items.

(i)	E
(ii)	D

2. Matching Items

(iii)	A
(iv)	C
(v)	B
(vi)	D
(vii)	B
(viii)	C
(ix)	A
(x)	E
@01 Marks = 10 Marks	

SECTION B (60 MARKS)

3. (a) Precaution for proper use of a microscope

LIST A	**LIST B**
(i)	F
(ii)	E
(iii)	G
(iv)	J
(v)	A
(vi)	K
(vii)	O
(viii)	C
(ix)	R
(x)	D
@01 Mark = 10 Marks	

i. Use both hands to hold the microscope, one hand should hold the arm and another hand should hold the base.
ii. Always place the Microscope on a desk or table carefully and gently.
iii. Always keep the microscope in an upright position. **@ 1Marks**
iv. Keep the Microscope covered when not in use.

(b)

A TEST EXPERIMENT	**CONTROL EXPERIMENT**
Is the scientific investigation procedure which is used to test or confirm a hypothesis **@ (1 Marks)**	Is necessary in scientific investigation because it differs from test experiment by only one factor thus confirming the conclusion.

4. (a) (i) Gaseous exchange refers to the exchange of gases between the body cell and the environment. **(1 MARKS)**

(ii) Nasal cavity → pharynx → Trachea → bronchi → bronchioles Alveoli **(1MARKS)**

(b) (i) Mammal → lungs
(ii) A young tadpole → external gills
(iii) An insect → Tracheal system **@1 MARKS**
(iv) Earthworm → whole body by diffussion
(v) Spider → lung books

5. (a) (i) A reflex action Is a simple form of automatic behavior in which the

stimulus
always bring about same response. **(1 MARKS)**

(ii) Examples of reflex actions are

- Moving the finger away from sharp object.
- Blinking of the eye. **@ 1 MARKS**
- Sneezing.
- Coughing. **TOTAL MARKS (04 MARKS)**
- Churning of food by stomach wall.

(iii) - Helps our body to carry out necessary activities without thought eg.digestion. **(01 MARKS)**

- Help an organism to escape from danger and ensuring the survival.

(01 MARKS)

(b) (i)

NERVOUS CONTROL		HORMONAL CONTROL	
(i)	Chemical and electrical impulse transmission in the nerve cell.	(i)	Chemical called hormone transmitted through blood stream.
(ii)	Rapid transmission and response.	(ii)	Slow transmission and relative slow response.
(iii)	It shows short term responses.	(iii)	It shows long term responses. **@ (01 MARKS)**

(b) (ii) A person was able to climb the tree quickly because of adrenaline hormone
secreted by the adrenal glands. This hormone stimulates the reactions that
prepare someone to run or fight an enemy. **(02 MARKS)**

1. (a) (i) **Gene Mutation:** These are usually spontaneous change in genes. **(01 MARKS)**

(ii) **Chromosomal Mutation:** These are changes that occur in structure of chromosomes. **(01 MARKS)**

(b) Four examples of genetic disorders caused by gene Mutation are:

i. Albinism.
ii. Sickle cell anaemia.
iii. Colour blindness.
iv. Haemophilia **@ 01 MARKS**
v. Down syndrome. **(TOTAL 04 MARKS)**
vi. Turner's syndrome.

2. (a) (i) **Transportation of Materials** Means movement of material from the environmet into the organisms or from one part of the organism to another or

from organism to the environment. **(01 MARKS)**

(ii) **Mass flow:** Is the bulk movement of substance from one region to another due to the difference in pressure between the two regions. **(01 MARKS)**

(b) A plant cell loses water, this cause the vacuole to shrink and the surface of the cell membrane to pull away from the cell wall making the cell flaccid or become plasmolysed by the process called plasmosis . **(02 MARKS)**

8. (a) (i) **Fertilization**: Is the fusion of the Male gamete with the female gamete to form a zygote. **(01 MARKS)**

(ii) **Ovulation**: I the liberation of an ovum from the ovary. **(01 MARKS)**

(b) There are two types of twins which are

i. Identical twins (Monozygotic twins) **(01 MARKS)**
ii. Fraternal twins (Non Identical twins) **(01 MARKS)**

FORMATION OF TWINS

(i) Identical twins are formed when a single egg is fertilized by one sperm, then the resulting zygote splits into two cells which develop independently. **(01 MARKS)**

(ii) Fraternal twins are formed when two ova are produced at the same time and both are fertilized by two different sperm. **(01 MARKS)**

9. (a) (i) Water, calcium, Potassium and sodium will be excreted by mammal in a greater amount under lower environment temperature because at lower temperature there is increase in Metabolic activity

(ii) Glucose will be excreted in urine under insufficient secretion of insulin. **(01 MARKS)**

(iii) More urea will be excreted under consumption of large amount of protein **(01 MARKS)**

(a) When a mammals body heat up the body regulates its temperature by sweating excessive. During this process water in form of sweat is lost from the body through evaporation. **(02 MARKS)**

10. (a) (i) **Genotype:** Is the genetic makeup of an organisms. **(01 MARKS)**

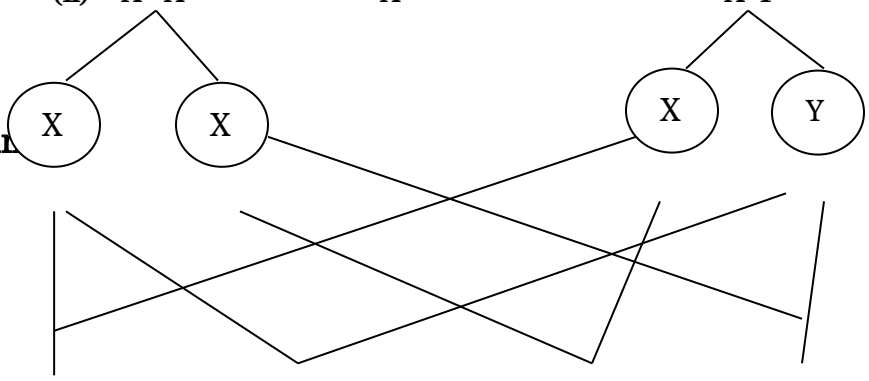
(ii) **Variation**: Is the possession of the characters which are different from those of their parents. **(01 MARKS)**

(iii) Phenotype: Is the physical appearance of an organisms. **(01MARKS)**

(b) (i) $X^H X^H$

(ii) $X^H X^H$ X X^hY (Parental genotype)

(Meiosis)



(Gamete) **(02 Mar**

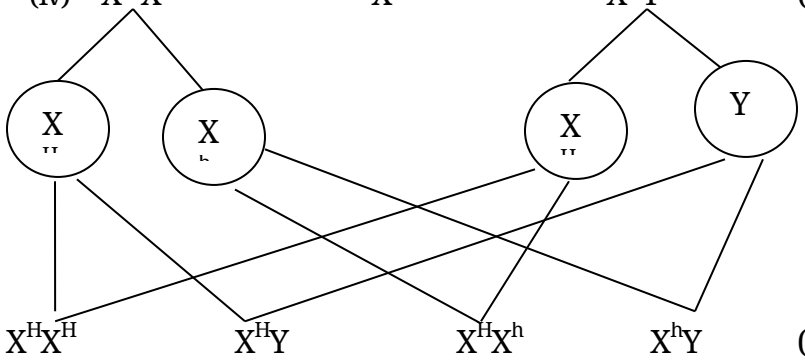
(Fertilization)

$X^H X^h$ X^HY $X^H X^h$ X^HY (off spring)

(iii) $X^H X^h$ (5) and X^HY (09). **(01 MARKS)**

(iv) $X^H X^h$ X X^HY (Parental genotype)

(Meisosis)



(Gamete)

02 MARKS

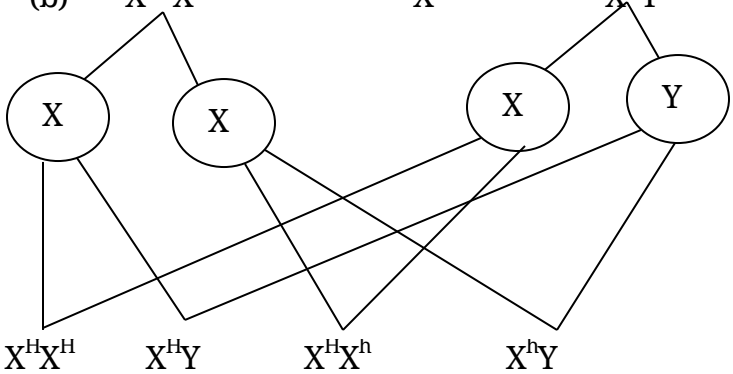
Fertilization

X^HX^H X^HY X^HX^h X^hY (off springs)

10. (b) $X^H X^h$ X X^HY (Parental genotype)

(Meiosis)

(Gamete)



fertilization

(02 MARKS)

X^HX^H X^HY X^HX^h X^hY (off springs)

Probability (P) = $\frac{\text{Number of event (E)}}{\text{Number of sample space (S)}}$

$P = \frac{1}{4} = 0.25 = 25\%$

They are 25% of getting haemophiliac son (01 MARKS)

SECTION C (20 MARKS)

11. **INTRODUCTION.**

- Food processing is the process of making food to be edible and palatable.
- Food preservation means the process used to prevent deterioration of food.
- Food storage means keeping reserves of food for the future use.

3 POINTS @ 1MARKS.

MAIN BODY

Traditional methods includes following

Smoking: This involve drying food using smoke to reduce the activity of Micro organisms.

Drying: Drying in the sun involves the removal of water from the food using solar energy to reduce the activity of Micro organisms.

Salting: This involve the addition of salt such as a common salt direct to food so as to remove water from the food.

Modern Methods also include the following:

Pasteurizing: This involve overheating the food above 55 - 60°C to destroy enzymes and Micro organisms.

Refrigeration: This is the storage method which involves keeping perishable food such as vegetable and milk.

Freezing: This involves keeping food like meat in a freezer at a temperature below 10°C so as to stop the growth of microorganism in the food.

Cunning and Bottling: In this method first the food is heated to kill microorganism then sealed in an air tight or bottle which does not allow oxygen to enter in the container.

ANY @ POINTS 2 $^1/_2$ MARK = 07 $^1/_2$ MARKS.

CONCLUSION

There are various methods of food preservation and storage which are very important so as to preventing growth and development of microorganisms. **@ 2 MARKS (GRAND TOTAL 20 MARKS)**

12. Six factors that affect the body immunity of the body in human being.

Introduction: Immunity is the ability of the body to resist infection and disease.

The factors that affect immunity include the following (03 Marks)

- **Age:** Young children and older people have reduced immunity, children they reduced immunity because they have not started producing antibodies in order to people the production of antibodies reduced as the body tissue age.
- **Food taken:** Some nutrients boosts body immunity for example Vitamin C which is naturally food in fruits such as oranges.
- **Types of disease:** Some diseases such as HIV/AIDS attack the body immune system.
- **Vaccination/immunization:** vaccination is the introduction of vaccine into the body. Therefore vaccine boosts body immunity. Also individual who are not immunized against are more likely to suffer from the disease.
- **Drugs:** Most of the drugs contain chemical substance that destroy tissue and weaker the body immunity. **@ (POINT O4 MARKS) TOTAL = 20 MARKS**.
- **Conclusion. (02 MARKS)**

13. The function of the major component of the human skeleton and their adaptions.

Introductions:

The human skeleton is divided into two major components which are axial skeleton and appendicular skeleton. The axial skeleton consists of the skull, vertebral column and rib cage while the appendicular skeleton consists of the pectoral and pelvic girdles and the bones of the fore and hind limbs. **(1 MARKS)**

Main body

(a) The function of the axial skeleton are protection, support and attachment of muscles

ANY 2 (A) 1 MARK = 2 MARKS

Adaptions

S/N	STRUCTURE	FUNCTIONS
(i)	Skull **(1 MARK)**	- It has a large space inside the cranium in which the brain resides. - Hard for protection. - Also for attachment of the neck and facial muscles Any 1 point = 1 Mark
(ii)	Vertebral column/vertebrae **(1 MARK)**	- Have prominent centrums for support of body weight and articulation with ribs. - Have a wide neural canal for passage of spinal cord. - Have wide spine which offers a large surface area for attachment of back muscels. - Have long large and well developed transverse processes for attachment of muscles that maintain posture and flex the spine. **ANY 1 POINT MARK = 1**

(iii)	Rib cages (Ribs) **(1 MARKS)**	- Have long shafts (bone) for attachment of intercostals muscles. - Have curved shafts to provide a large surface area for attachment of intercostals muscles. - Have hard shaft for support and protection of delicate organs of thoracic cavity from mechanical damage. **ANY 1 POINT (1 MARKS)**

TOTAL 6 MARKS

(b) The appendicular skeleton which consists of the upper lower extremity including the pectoral and pelvic girdles and the bones of the fore and hind limbs.

The functions of the appendicular skeleton are to **provide base for articulation** and **attachment of muscles** and also for **movement.** Any two points **@ (1 MARKS = 2 MARK)**

ADAPTIONS

S/N	STRUCTURE	ADAPTATIONS
(i)	Pectoral girdle **(1 MARKS)**	- Consist of hattened bones which provide large surface area for muscle attachment.
(ii)	Pelvic girdle **(1 MARKS)**	- Consists of flattened bones which provide large surface area for muscle attachment. - Provides base for articulation with high bones. - Have ring hole which allow free movement of bones which assist delivery. **ANY 1 POINT = 1 MARK**
(iii)	Bones of fore and hind limbs **(1 MARKS)**	- Bones of fore limbs have dongated bones with ample joints. The bone provides proved attachment of fore limb muscles. - Bones of the hindlimbs also are elongated bones with ample joints and to provide attachment for hind limbs muscles. **ANY POINTS = 1 MARKS**
(iv)	Joints **(1 MARKS)**	- They are filled with synovial fluid which acts as lubrication. - The articulating surface is covered with cartilage which absorbs shock. - They are prevented from coming apart by ligament. **ANY POINT = 1 MARK**

TOTAL 8 MARKS

CONCLUSION

The main functions of the human skeleton are support, movement, locomotion, protection and production of blood cells. Human skeleton is adapted to bring about movement by having joints and protection against compression. **1 MARKS**

GRAND TOTAL 20 MARKS.

BISMILLAH RAHMAN RAHIIM
THE NATIONAL MUSLIM COUNCIL OF TANZANIA
ZAIM EDUCATION DEVELOPMENT

FORM FOUR ZAIM SCHOOLS MOCK EXAMINATION

031/1 **PHYSICS 2**
MARKING SCHEME

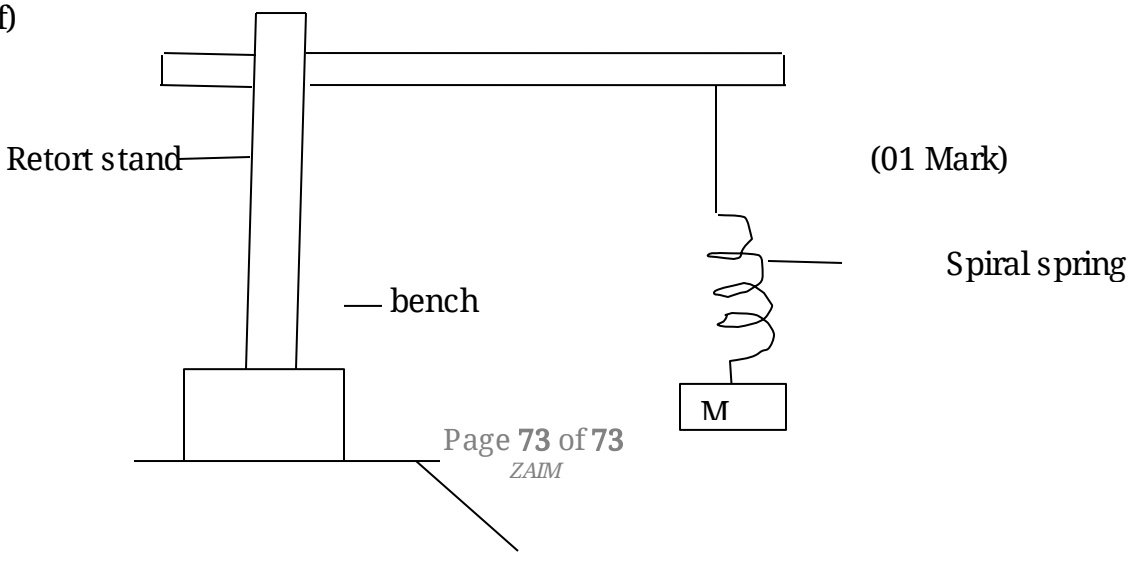
Time: 3 Hours **Saturday, 29th June 2019 a.m.**

1. Table of results.

(e) TAKE $1\frac{1}{2}$ MARKS IN EACH

M(g)	T(s)	T(s)	Tc^2 S^2
50	9.6	0.48	0.23
100	13.2	0.66	0.43
150	15.8	0.79	0.63
200	18.2	0.91	0.83
250	20.2	1.01	1.03

(f)



Ground floor

(g) (ii) Points (225,0.9315) (1/2Marks) (87.5, 0.345)

Slope = $\frac{\Delta T^2}{\Delta M(g)}(S^2) = \frac{0.9315-0.345}{225-87.5} = 0.0043$ (1/2 Mark)

Slope = $0.0043(S^2/g)$ (Allowable range 0.0039 – 0.0045) (01 Marks)

(iii) From

$T^2 = 4\pi^2\left(\frac{m+s}{k}\right)$

$T^2 = \frac{4\pi 2}{K} m + \frac{4\pi^2}{k} S$ (1/2 Marks)

Since slope = $\frac{4\pi 2}{K}$ (1/2 Marks)

Make K the subject

$K = \frac{4\pi 2}{\text{Slope}} = \frac{4\pi 2}{0.0043} = 9181.03 \frac{g}{S^2}$ (Allowable Range $9000\frac{g}{S^2}$ - $9900\frac{g}{S^2}$

OR $90\frac{kg}{S^2}$ - $99\frac{kg}{S^2}$ (2 Marks)

To set S from the graph – the value of T^2 intercept = 0.03 (1/2 Marks)

Then $\frac{4\pi 2}{K} S = 0.035^2$ (1/2 Marks)

Make S the subject

(Allowable range S = 6 – 8g) $S = \frac{0.03}{4\pi 2}$ (1/2 Mark) $= \frac{0.035^2 \times 9181.03}{4\pi^2} \frac{g}{S^2} = 7.5$ (1/2Marks)

= **The value of S is 7.5g (1/2 Marks) Answer**

(h) (i) Source of error (Note: errors are so many)

(a) Incorrect reading of stop watch e.t.c (1/2 Marks)	Correct reading e.t.c (1/2Marks)

(ii) The aim of this experiment is to determine spring constant K and effective mass of spring. (01 Marks).

GRAPH

2. **(B)** (iii) Table of results.

TAKE 01 MARKS EACH

R(∩)	I (A)	V(v)
1	1.5	1.5
3	0.8	2.2
5	0.5	2.5
7	0.4	2.6
9	0.3	2.7

(C) (ii) Points (0.2 , 2.8) (1.4 , 1.6)

$$\text{Slope} = \frac{\Delta V}{\Delta I}(v) = \frac{1.6 - 2.8}{1.4 - 0.2} = -1 \quad (01 \text{ Marks})$$

Slope -1 v/A (1/2 Marks)

(iii) The physical meaning of slope is internal resistance. (02 Marks)

(D) From $V = -Ir + E$ compare with equation of straight line

$Y = xm + c$

Since slope = internal resistance r ∴ r = 1 (02 Marks)

And E = V – intercept and from the graph

V – intercept 3V ∴ E = 3V (02 Marks)

(E) The aim of this experiment is to determine the e.m.f of dry cells and internal resistance (02Marks)

(F) Source of errors

(i) Zero error (01 Marks)

(ii) Incorrect readings of ammeter and voltimeter (01 Marks)

Precautions/ways of minimizing

(i) Ensure the ammeter and voltimeter start from zer mark.

(ii) Ensure correct reading of the ammeter and voltimeter (01 Marks)

NOTE: Source of errors are so many.

GRAPH

BISMILLAH RAHMAN RAHIIM
THE NATIONAL MUSLIM COUNCIL OF TANZANIA
ZAIM EDUCATION DEVELOPMENT

FORM FOUR ZAIM SCHOOLS MOCK EXAMINATION

BASIC MATHEMATICS
(For Both School and Private Candidates)

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1. Answer
   (a) Soln
   Round off 4.1 to one
   4.1≈4.0
   Round off 0.082 to hundredths
   0.082 ≈ 0.08
   4.1 x 0.082 ≈ 4.0 x 0.08 = 0,32
   :. 4.1 x 0.082 ≈0.32

   (b) Soln
   i. Let

X = 0.83……..(i)

10x = 8.33 ……(ii)

Take equation (ii) – equation (i)

10x – x = 8.33 – 0.83

$\frac{9x}{9} = \frac{7-5}{9}$

$X = \frac{75}{90}$

$X = \frac{5}{6}$

.: $0.83 = {}^{5}/_{6}$

ii. Soln

$\frac{125}{1000} \times \frac{1}{100} = \frac{125}{100,000}$

$= 1/800$

:. $0.125^{\%} = {}^{1}/_{800}$

2. Answer

(a)

i. Soln

$({}^{1}/_{3})^{x} = \frac{1}{124}$

$({}^{1}/_{3})^{x} = \frac{1}{5^{5}}$

$({}^{1}/_{3})^{x} = ({}^{1}/_{5})^{5}$

.: X 5

2.

ii. $\frac{8x^{2}}{8} = \frac{288}{8}$

$X^{2} = 36$

$\sqrt{2}x = \sqrt{36}$

:. X 6

(b) Soln

Logy y + logx2 = 3
$Log_{10}(yx^2)$ = 3

yx^2 = 10^3

yx2 = 1000
x2 = x2

:. y = $1000 / x^2$

3. Answers
   (a) Soln
   Let a = 2
   b= y
   2*y = 2^2 + y
   2*y = 4 = y
   4*4+y 25

   Let a = 4
   b = 4+y

   4*4 + y = 4^2 + 4 +y
   25 = 14 + 4 + y
   y =25 – 20
   :.Y = 5

   (b) Soln
   A= {1,2,3}
   s = 2n
   s=23
   s=8

   A = { }, {1}, {2}, {3}, {1,2}
   {1,3}, {2,3}, {1,2,3}

4. Answer
   (a) Soln
   Slope oa AB = $\frac{6-2}{5+3} = \frac{4}{8} = \frac{1}{2}$
   Slope of BC = $\frac{2-6}{} = \frac{-4}{} = -2$

$$7 - 5 \qquad 2$$

$M_{AB} \times M_{BC} = -1$

$\frac{1}{2} \times -2 = -1$

$-1 = -1$

:. Since the product of their slope get -1, then triangle ABC is right angled at B

(b) Soln

$U = \sqrt{12^2+5^2}$

$= \sqrt{144+25}$

$= \sqrt{166}$

13

$U = U = \frac{1}{3}(12, 5) = (\frac{12}{12}, \frac{5}{13})$

:.U = $\frac{12i}{13} + \frac{5i}{13}$

5. Answer

(a) Soln

Since $\overline{ZY}$ II $\overline{MN}$ then

$XYZ = XMN$ (Corresponding anlgels)

$XYZ = XMN$ (Corresponding anlgel)

Hence

$\nabla XYZ \sim \nabla XMN$ (AA – Similarity theorem)

:. $\frac{XY}{XM} = \frac{XZ}{XN} = \frac{YX}{MN}$

$\frac{10cm}{XM} = \frac{9cm}{12cm} = \frac{6cm}{MN}$

$\frac{10cm}{XM} = \frac{9cm}{12cm}$

$\overline{XM} = {}^{13}/_{12cm}$

But

$\overline{MY} = \overline{XM} - \overline{XY}$

$$MY = \frac{13\frac{1}{3}}{3\frac{1}{3}} - 10cm$$

$$\frac{6\ cm}{NM} = \frac{9\ cm}{12\ cm} \qquad NM = 8\ cm$$

:. MY $13\ ^{1}/_{3}$ and NM = 8 cm

5 Answer

(b) Soln

$$P = 2nr\ Sin\ \frac{180°}{n}$$

Given

P = 50 m

n = 6

$50 = 2 \times 6 \times r \times Sin\ 30^{0}$

$$r = \frac{50\ m}{12 \sin m}$$

r = 8.33m

:. The radius is 8.33m

6. Answer

(a) P & $\frac{1}{Q2}$

$$p = \frac{K}{Q2}$$

$K = PQ^{2}$

$K = 4 \times 8^{2} = 4 \times 64$

K = 256

$$Q^{2} = \frac{K}{P}$$

$$Q = \sqrt{\frac{256}{15}}$$

Q = $\sqrt{16}$

Q = 4

:. Q = 4 if P = 16

(b) Soln

6km = 1 litre

960 km = x

$= \frac{960 \text{ km} \times 1 \text{ Litre}}{6\text{km}}$

= 160 Litre

1 Litre = sh 2500

160 Litre = ?

= 160L x 2500

= 400,000

The cost of diesel Tsh. 400,000?=

= 2500 x 1600

= 4,000,000

:. The Cost of Diesel Tshs 4,000,000/=

7. Answer

(a) Soln

6 + 4 + 5 = 15

$X = \frac{6}{15}$ x 30,000 = sh 120,000/=

$Y = \frac{4}{15}$ x 30,000 = sh 80,000/=

$Z = \frac{5}{15}$ x 30,000 = sh 100,000/=

(b) Soln

100% = X

70% = 490,000

$$\frac{70\%}{70\%} = \frac{100\% \ X \ 490{,}000}{70\%}$$

X = 700,000/=

:. The Price is Tshs 700,000/= before reduction

8. Answer

(a) Soln

Given

$A_1 = {}^{-}8$

$A_n = 82$

$d = {}^{-}2 + {}^{+}8 = 6$

from

$Sn = \frac{n}{2}(A_1 + A_n)$

But $An = A1 + (n - 1)d$

$82 = -8 + (n - 1)\,6$

$82 = -8 + 6n - 6$

$82 = 6n - 14$

$$\frac{96}{6} = \frac{6n}{6}$$

$N = 16$

From

$$Sn = \frac{n}{2}[2A, + (n\text{-}1)d]$$

$$= \frac{16}{2}[2(-8), + (16\text{-}1)\,6]$$

$S_{16} = 596$

:. The sumu of series 592

8.

(b) Soln

$G5 = -2$

$G7 = -18$

$G5 = G_1r^4$

$G_1r^4 = -2$ ……(i)

$G_7r^4 = G_1r^6$

$G_1r^6 = -18$ .....(ii)

From Equation (i)

$G1 = \frac{-2}{r4}$ .....(iii)

Substute equation (iii) into equation (ii)

(i) $\frac{-2r6}{r4} = -18$

$\sqrt{r^2} = \sqrt{9}$

:. $r = 3$

(ii) Soln

$Sn = G1 \left(\frac{rn-1}{r-1}\right)$

$G1 = \frac{-2}{r^4} = \frac{-2}{3^4} = \frac{-2}{81}$

$G1 = \frac{-2}{81}$

$S_{10} = \frac{-2}{81}\left[\frac{3^{10}-1}{3-1}\right]$

9. Answer.

(a) Soln

Cos 135 = Cos $(45^0 + 90^0)$

Cos $(45^0 + 90^0)$ = Cos 45 Cos 90 – Sin 45 Sin 90

$= \left(\frac{\sqrt{2}}{2}\right)(0) - \left(\frac{\sqrt{2}}{2}\right)(1)$

Cos 135 $= -\frac{\sqrt{2}}{2}$

Tan $60^0 = \frac{\text{Sin } 60}{\text{Cos } 60}$

$\frac{\sqrt{3}}{2} \times \frac{1}{2} = \sqrt{2}$

Tan 60^0 = $\sqrt{2}$

Sin 450 $= \frac{\sqrt{2}}{\sqrt{2}}$

Sin (135) = Sin 45 Cos 90 + Cos 45 Sin 90

$\left(\frac{\sqrt{2}}{2}\right)(0) + \left(\frac{\sqrt{2}}{2}\right)(1)$

$= \frac{\sqrt{2}}{2}$

then $-\frac{\sqrt{2}}{2} + \left(\sqrt{3} \quad \frac{\sqrt{2}}{\sqrt{2}}\right)$

$\frac{\sqrt{2}}{\sqrt{2}} \cdot \frac{1}{2}$

$=- \frac{\sqrt{2}}{2} + \frac{\sqrt{6}}{2}$

$\frac{\sqrt{2}}{4}$

$= \frac{\sqrt{6} - \sqrt{2}}{2} \div \frac{\sqrt{2}}{4}$

$= \frac{\sqrt{6} - \sqrt{2}}{2} \times \frac{4}{\sqrt{2}}$

$= \frac{4\sqrt{6} - 4\sqrt{2}}{2\sqrt{2}}$

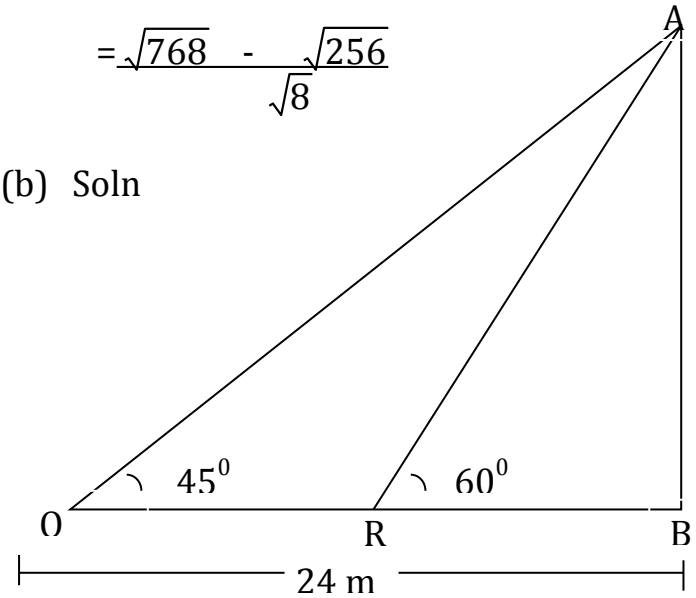
$= \frac{\sqrt{16 \times 6} - \sqrt{16 \times 2}}{\sqrt{4 \times 2}}$

$= \frac{\sqrt{96} - \sqrt{32}}{\sqrt{8}} \quad X \quad \frac{\sqrt{8}}{\sqrt{8}}$

$= \frac{\sqrt{96 \times 8} \quad \sqrt{32 \times 8}}{\sqrt{8}}$

$= \frac{\sqrt{768} - \sqrt{256}}{\sqrt{8}}$

(b) Soln



Let the height of Flag Staff AB

$\frac{AB}{} = \tan 45 = 1$

$24\overline{m}$

$\overline{AB}$ = 24m.

By triangle ARB

$\frac{\overline{AB}}{RB} = \tan 60 = \sqrt{3}$

$\frac{\overline{RB}}{\overline{AB}} = = \frac{1}{\sqrt{3}} = \frac{\sqrt{6}}{2}$

$\frac{\overline{RB}}{24M} = \frac{\sqrt{3}}{3}$

$\frac{\overline{RB}}{24M} = \frac{\sqrt{3}}{3}$

$\overline{RB} = \frac{24\sqrt{3}}{3} = 8\sqrt{3}$

$\overline{PR} = 24 - 8\sqrt{3}$

$= 8(3-\sqrt{3})$ m

=PR = 10.144m

:. PR = 10.144m

10. ANSWERS

(a) Soln

$7^2X^2 - 5^2 = 0$

(7x)2 - 52 = 0

(7x - 5) (7x - 5) = 0

7x = 5 or 7x = -5

:. X = $^5/_7$ or X = $^{-5}/_7$

(b) Soln

$1350\left(\frac{x+3}{2}\right) - 1350x = \frac{5}{1}$

$$\frac{1350x + 4050 - x}{x^2 + 3x} = \frac{5}{x1}$$

$4050 = 5x^2 + 15x$

$5x^2 + 15x = 4050$

$x^2 + 3x = 810$

$x^2 + 3x - 810 = 0$

$x^2 + 30x - 27x - 810 = 0$

$x(x+30) - 27(x + 30) = 0$

$(x+30)(x - 27) = 0$

$x = -30$ or $x = 27$

$\therefore x = 27$

SECTION B

11. ANSWER

Soln

	Assembly	**Finishing**
Table	3 x	x
Chair	48	2y
	48	36

Let X – be the number of table

Y – be the number of chair

Forming Inequalities

$3x + 4y = \leq 48$ …… (i)

$x + 2y = \leq 36$……...(ii)

$x \geq 0 = y \geq 0$

Objective function

$F(x, y) = 50x + 40y$

Corner Points	**Objective**	**Maximum**
	50 x + 40 y	
0(0, 0)	0 + 0	0
A(0, 12)	0 + 480	480
B(8, 6)	400 + 240	640
C(12, 0)	480 + 0	480

The Point B(8, 6) offers the best Combination of chairs and tables to be produced. So as to maximize profit that is, 8 tables and 6 chairs have to be produced.

12. Answers

(a) Soln

A = 4

Class boundary	**Class Mark (x)**	**Frequency**	**Comm. free**	**d – x - A**	**Fd**
21 - 30	25.5	6	6	21.5	129
31 - 40	35.5	6	12	31.5	189
41 - 50	45.5	5	17	41.5	207.5
51 - 60	55.5	4	21	51.5	206
61 - 70	65.5	13	34	61.5	799.5
71 - 80	75.5	8	42	71.5	572
80 - 90	85.5	1	43	81.5	81.5
91 - 100	95.5	2	45	91.5	186.0
		45			**2367.5**

12.

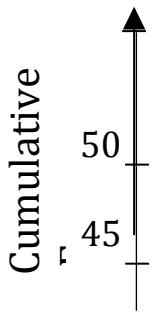
(b) $\bar{X} = A + \frac{\sum fd}{\sum f} =$

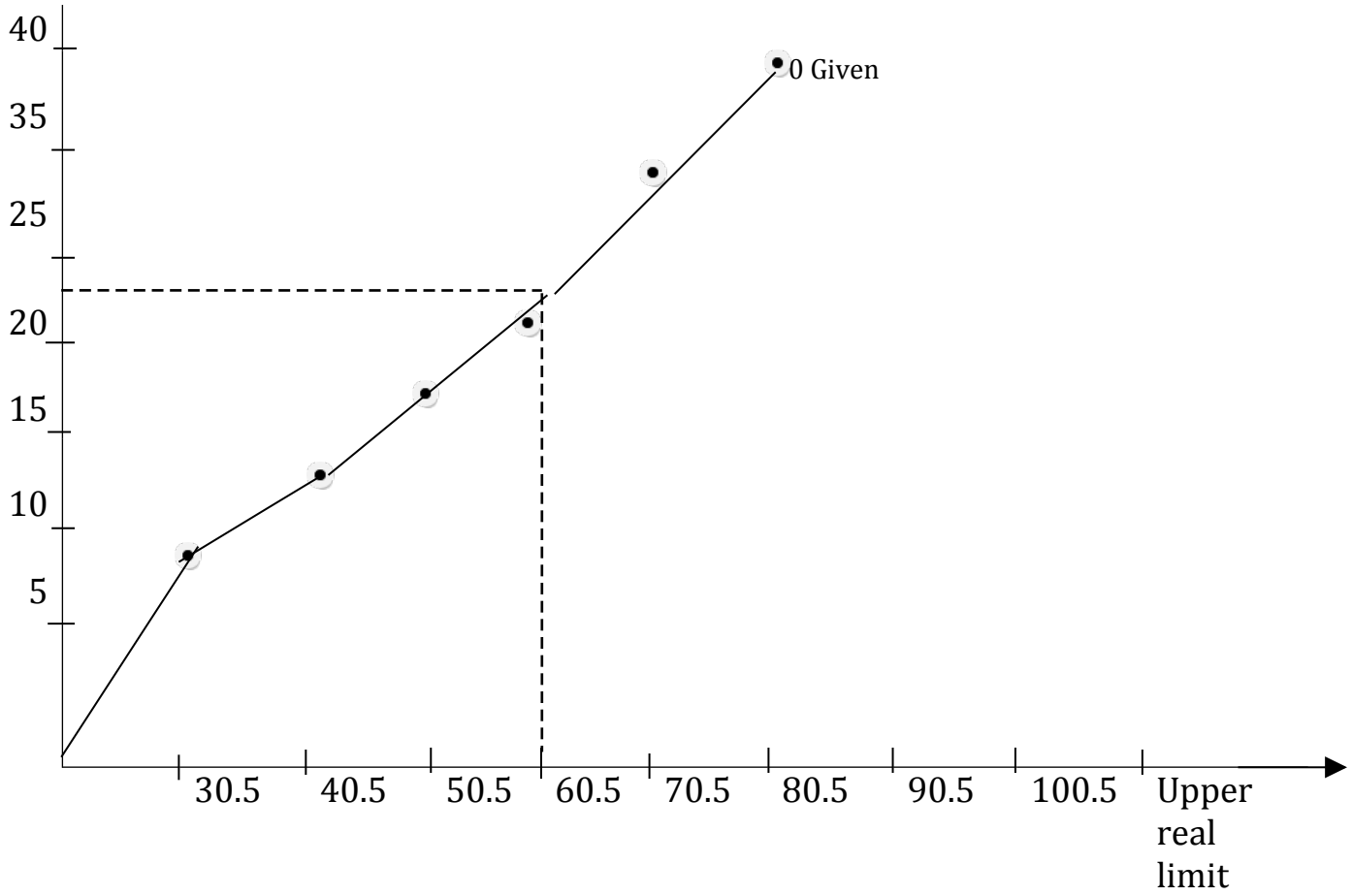
$= 4 + \frac{2367.5}{45}$

$\bar{X} = 4 + 52.61$

:. $\bar{X} = 56.61.$

(c) ANSWER





From the graph

:.Media is 60.5

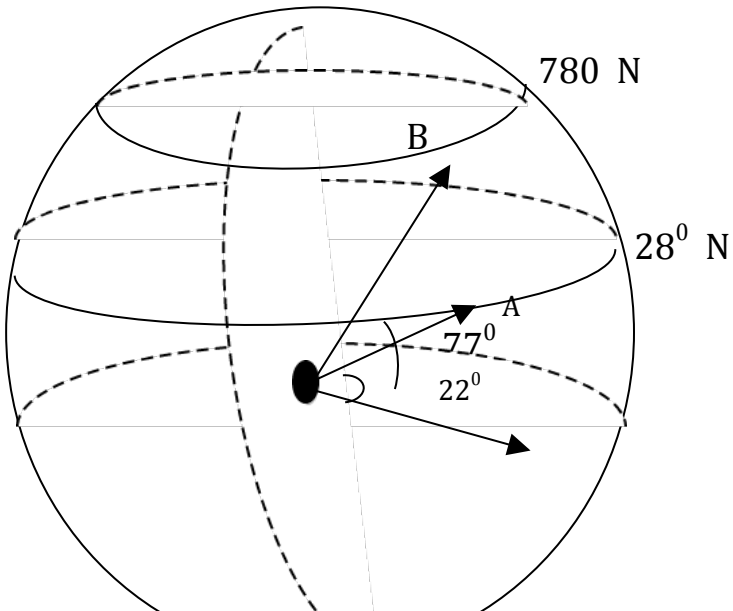
77^0

22^0

13. Answers

(a) Soln

i. Answer





Since both places A an B are on the North of the Equator

The central angel subtended by are
Ab,(Q) = 78^0 - 18^0 = 50^0
Q = 50

ii. Q = 64^0 - 44^0
:.Q = 20^0

(b) Soln
Given
$\overline{OA}$ = OC = r = 5cm
$\overline{PC}$ = *8cm*

Then, PO = OC + PC = 5cm + 8cm

:. $\overline{PO}$ = 13cm

Thus, by applying Pythagoras theorem in

ΔPAO: $\overline{OP^2} = \overline{OA^2} + \overline{AP^2}$

$13^2 = 52 + \overline{AP^2}$

$\sqrt{AP^2} = \sqrt{144}$

AP = 12CM

The Length of the tangent is 12 cm

14. Soln

Cash account (1)

Dr						**Cr**	
Date	Particular	F	Amount	Date	Particular	F	Amount

Feb . 1	Capitol	2	60,000	Feb. 2	Purchase	3	50,000
Feb . 10	Sales	5	90,000				
				Feb . 3	Transport	4	10,000
				Feb 20	Furniture	6	30,000
				Feb. 28	Carriage	7	5,000
				Feb. 28	**Balance**	**c/d**	**55,000**
			150,000				**150,000**
Mar. 1	**Balance**	**b/d**	**55,000**				

CAPITOL ACCOUNT (2)

Dr Cr

Date	Particular	F	Amount	Date	Particular	F	Amount
		c/d	60,000		Cash		60,000
			60,000				
					Balance	**b/d**	**60,000**

PURCHASES ACCOUNT (3)

Dr **Cr**

Date	Particular	F	Amount	Date	Particular	F	Amount
	Cash		50,000		Balance	c/d	50,000
			50,000				**50,000**
	Balance	**b/d**	**50,000**				

TRANSPORT ACCOUNT (4)

Dr **Cr**

Date	Particular	F	Amount	Date	Particular	F	Amount
	Cash		10,000		Balance	c/d	10,000
			10,000				**10,000**
	Balance	**b/d**	**10,000**				

SALES ACCOUNT (5)

Dr **Cr**

Date	Particular	F	Amount	Date	Particular	F	Amount
	Balance	c/d	90,000		Cash		90,000
			90,000				
					Balance	**b/d**	**90,000**

14. question

FURNITURE ACCOUNT (6)

	Dr				Cr		
Date	Particular	F	Amount	Date	Particular	F	Amount
	Cash		30,000		Balance	c/d	30,000
			30,000				**5,000**
	Balance	**b/d**	**30,000**				

FURNITURE ACCOUNT (7)

	Dr				Cr		
Date	Particular	F	Amount	Date	Particular	F	Amount
	Cash		5,000		Balance	c/d	5,000
			5,000				**5,000**
	Balance	**b/d**	**5,000**				

TRIAL BALANCE AS AT 28TH FEBRUARY

S/NO	PARTICULAR	FOLIO	DR	CR
1.	Cash in hand	1	55,000	
2.	Capital	2		60,000
3.	Purchases	3	50,000	
4.	Transport	4	10,000	
5.	Sales	5		90,000
6.	Furniture	6	30,000	
7.	carriage	7	50,000	
			150,000	150,000

15. Answer

(a) Soln

$$\begin{pmatrix}3 & 2\\4 & 5\end{pmatrix}\begin{pmatrix}x\\y\end{pmatrix}=\begin{pmatrix}12\\23\end{pmatrix} \text{.....(1)}$$

Let A = $\begin{pmatrix}3 & 2\\4 & 5\end{pmatrix}$

Find (A) = 15 - 8 = 7

Then find

$$A^{-1} = \frac{I}{[A]}\begin{pmatrix}5 & -2\\-4 & 3\end{pmatrix}$$

$$= \frac{1}{[7]}\begin{pmatrix}5 & -2\\-4 & 3\end{pmatrix}$$

$A^{-1} = \begin{pmatrix} 5/7 & -2/7 \\ -4/7 & 3/7 \end{pmatrix}$

Multiply Inverse to both side of equation (i)

$$\begin{pmatrix} \frac{5}{7} & -\frac{2}{7} \\ -\frac{4}{7} & \frac{3}{7} \end{pmatrix}\begin{pmatrix} 3 & 2 \\ 4 & 5 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \frac{5}{7} & -\frac{2}{7} \\ -\frac{4}{7} & \frac{3}{7} \end{pmatrix}\begin{pmatrix} 12 \\ 23 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \frac{60}{7} & + & -\frac{46}{7} \\ -\frac{48}{7} & + & \frac{69}{7} \end{pmatrix}$$

$$\begin{pmatrix} x & + & 0 \\ y & + & 1 \end{pmatrix} = \begin{pmatrix} \frac{14}{7} \\ \frac{48}{7} \end{pmatrix}\begin{pmatrix} \frac{14}{7} \\ \frac{48}{7} \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$$

$\therefore (x, y) = (2, 3)$

(b) Soln

$y = 2x + 5$

$\& = 45^0$

Matrix

$$M = \begin{pmatrix} \cos 90 & \sin 90 \\ \sin 90 & -\cos 90 \end{pmatrix}$$

$$\mathbf{M} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

15. answer

Let P1 $(x_1\ y_1)$ and Q1$(x_2\ y_2)$ be the image of P(0,5) and Q$(^{-5}/_{21}\ 0)$

$$\begin{pmatrix} x1 \\ y1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 0 \\ 5 \end{pmatrix} = \begin{pmatrix} 5 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} x2 \\ y2 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} -\frac{5}{2} \\ 5 \end{pmatrix} = \begin{pmatrix} 0 \\ -\frac{5}{2} \end{pmatrix}$$

From $\frac{y - y1}{x-x1} = \frac{y2 - y1}{x2-x1}$

$$\frac{y - 0}{x-5} = \frac{0 - (-\frac{5}{2})}{5-0}$$

$$y = \frac{1}{2}x - \frac{5}{2}$$

:. The image of $4y = 2x + 10$ after reflection in the line $y = x$

$$y = \frac{1}{2}x - \frac{5}{2}$$

15.

(c) Soln

$$\begin{pmatrix} x1 \\ y1 \end{pmatrix} = \begin{pmatrix} \cos 90 & \sin 90 \\ \sin 90 & -\cos 90 \end{pmatrix} = \begin{pmatrix} -1 \\ 6 \end{pmatrix}$$

$$= \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} -1 \\ 6 \end{pmatrix}$$

$$\begin{pmatrix} x1 \\ y1 \end{pmatrix} = \begin{pmatrix} 0 & -6 \\ -1 + 0 \end{pmatrix} = \begin{pmatrix} -6 \\ -1 \end{pmatrix}$$

$$\begin{pmatrix} x1 \\ y1 \end{pmatrix} = \begin{pmatrix} \text{Cos}180 & -\text{Sin } 180 \\ \text{Sin } 180 & \text{Cos } 180 \end{pmatrix} = \begin{pmatrix} -6 \\ -1 \end{pmatrix}$$

$$\begin{pmatrix} xii \\ yii \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} -6 \\ -1 \end{pmatrix}$$

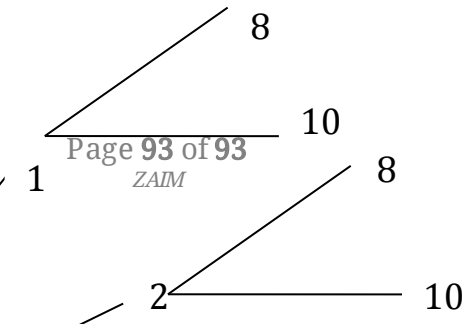
$$\begin{pmatrix} 6 + 0 \\ 0 + 1 \end{pmatrix}$$

$$(xii, yii) = (6, 1)$$

:. R_{180} R_{90}= $(-1, 6) = (6, 1)$

16. Answers

(a) **Question**



(i) **Soln**

(ii) $S = \left[\frac{1}{8} \quad \frac{1}{10} \quad \frac{2}{8} \quad \frac{2}{10} \quad \frac{4}{8} \quad \frac{4}{10} \quad \frac{6}{8} \quad \frac{6}{10}\right]$

$E = \left[\frac{1}{8}, \frac{1}{10}, \frac{2}{8}, \frac{2}{10}, \frac{4}{8} \quad \frac{4}{6}\right]$

$n(E) = 5$
$n(S) = 8$

$P(E) = \frac{n(E)}{n(S)}$

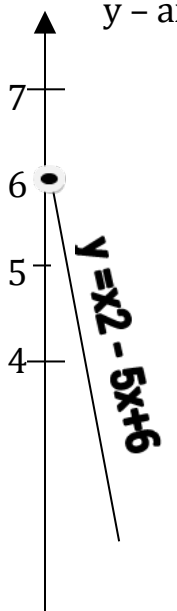
$= \frac{5}{8}$

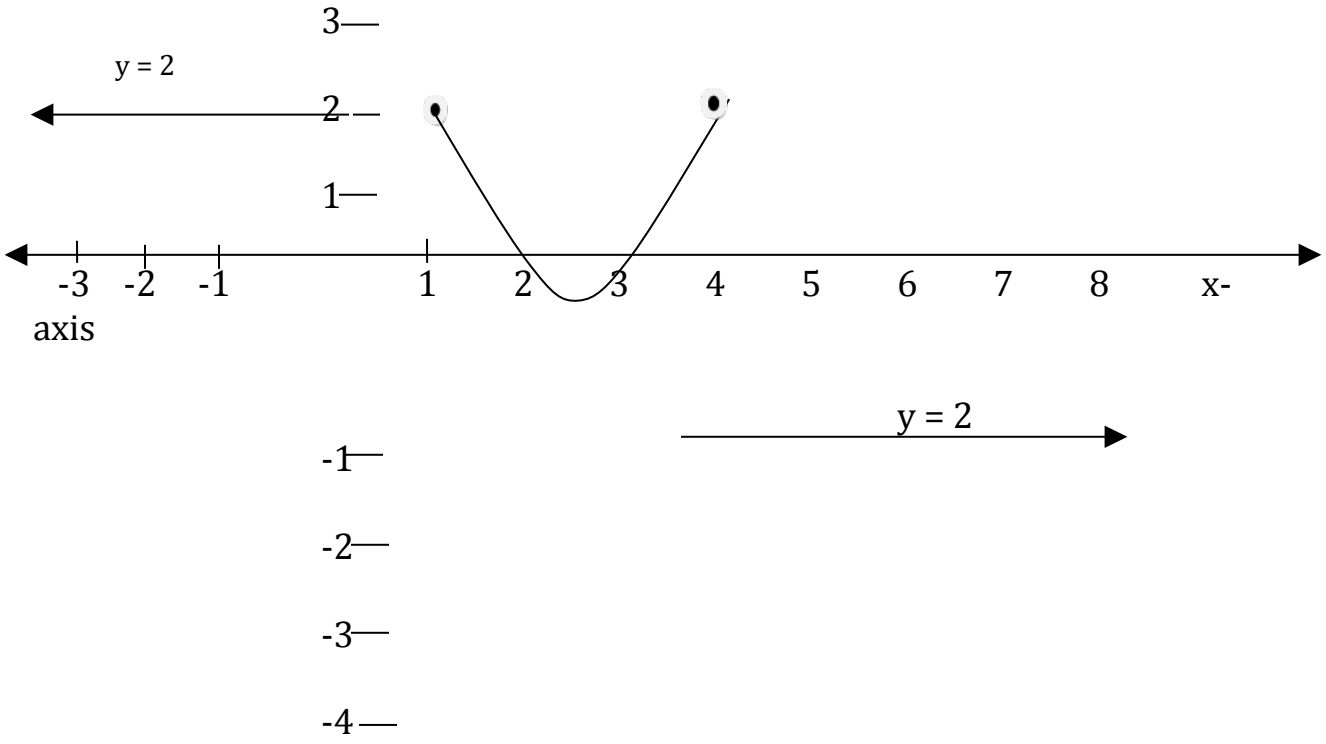
:. Probability is $\frac{5}{8}$

(b) ANSWER

(i) Soln

x	0	1	2	3	4
y =x2 - 5x+6	6	2	0	0	2





(ii) Domain = $[x:x \geq 0]$

Range = $\left[y:y= -2 \text{ and } \frac{-4}{4} \leq 6\right]$

(iii) $F(5) = -2$

$F(0) = 6$

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ZAIM EDUCATION DEVELOPMENT LIMITED

FORM FOUR ZAIM SCHOOLS MOCK EXAMINATION

011 **KISWAHILI**
MARKING SCHEME

SEHEMU A

1. (a) Maana ya maneno yaliyokolezwa wino

Mlimta – kumpenda sana (Alama $\frac{1}{2}$)

Chakalamu – kupagawa (Alama $\frac{1}{2}$)

Vimwana – wanawake (Alama $\frac{1}{2}$)

Mhemko - hisia za kimapenzi (Alama $\frac{1}{2}$)

Jau - shida, tabu, tatizo n.k

(b) Jina la shairi Moyo (Alama 1)

(c) Mashaka aliyonayo mwandishi juu ya mwenendo wa moyo ni kitendo cha moyo kuwaani wanawake kwa kiasi kikubwa huku akisahau kwamba hao wanawake walishawahi kumfanyia kitu kibaya (Alama 1)

(d) Katika jamii ya leo mapnzi yamekuwa yakiwatesa watu wengi, mapenzi yamepelekea kutokuelwana baina ya watu kutokana na kutokuwa na uaminifu miongoni mwa wapenzi (Alama 1)

2. Mtahiniwa anatakiwa kuandika ufupisho kwa kuzingatia taratibu za uandishi wa ufupisho.
   i. Kichwa cha ufupisho wa habari, kiwe katikati juu na kiandikwe kwa herufi kubwa na kipigiwe mstari. (Alama1)
   ii. Mantiki na mtiririko wa mawazo (Alama 1)
   iii. Kutumia maneno yake mwenyewe bila kupotosha mantiki ya habari chanzi (Alama 1)
   iv. Kuandika kwa muktasari kadri iwezekanavyo bila kuacha maana ya msingi ya habri chanzi (Alama 1)
   v. Kuzingatia idadi ya maneno (Alama 1)

SEHEMU B (ALAMA 25)
SARUFI NA UTUMIZI WA LUGHA
Jibu maswali yote

3. Mwanafunzi anatakiwa aonyesha dhima tano, atoe mifano miwili kwa kila dhima

   - Kutaja dhima (Alama $\frac{1}{2}$)
   - Kila mfano (Alama $\frac{1}{4}$)

   i. Kuonyesha nafsi
   Mfano Ninasoma
   Ni - hudokeza nafsi ya kwanza umoja
   Tunasoma – hudokeza nafsi ya kwanza wingi

   ii. Kuonyesha wakati/ njeo
   Mfano anayekula – huonyesha wakati uliopo
   Watakaokula – huonyesha wakati ujao

   iii. Kuonyesha urejeshi
   Mfano anayecheza
   Wanaoimba

   iv. Kuonyesha hali mbalimbali za vitenzi
   Mfano hucheza – hali ya mazoea
   Hapendi – hali ya ukanushi

   v. Kuonyesha umoja na wingi

Mfano kiti – viti

vi. Kuonyesha kauli mbalimbali za vitenzi.

4. Mwanafunzi anatakiwa aoneshe faida za michoro katika kamusi. Alama 1 kwa kila faida.
   i. Huvuta nadharia ya watumiaji kamusi na kuwafanya wazingatie wanachotazama
   ii. Huwawezesha watumiaji kamusi . kuunda dhana ya jambo, hasa kama ni geni
   iii. Husaidia watumiji kamusi. Kutofautisha vitu vinavyokaribiana kimaoni au kimaumbile
   iv. Huwawezesha watumiaji kamusi. Kuona mfano baina ya vitu
   v. Maumbo hunata katika kumbukumbu za watumiaji na huwawezesha kukumbukamaana kwa urahisi.
   vi. Uhitalitiano huifanya kamusi ivutie hasa zinapotumia rangi.

5. (a) Mwanafunzi anatakiwa aonyeshe tofauti kati ya kishazi na sentensi kwa mifano (Alama $2\frac{1}{2}$)

   Kishazi ni tungo yenye kitenzi ambacho chaweza kujitosheleza na kukamilisha ujumbe uliokusudiwa na mzungumzaji au kuwa na kitenzi kisichoweza kujitegemea chenyewe. (Alama 1)

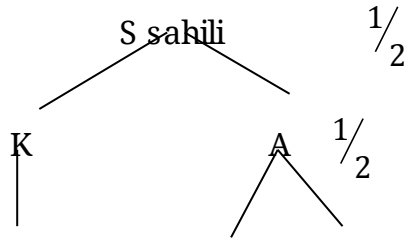
   Mfano Mtoto aliyepotea jana. (Alama $\frac{1}{4}$)

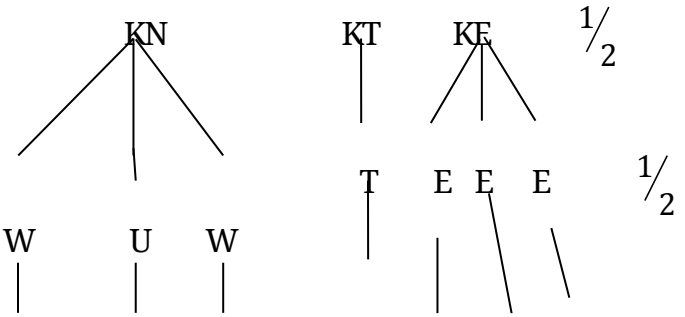
   Sentensi ni kifungu cha maneno kuanzia neno moja na kuendelea chenye kiima na kiarifu na inaleta maana kamili . (Alama 1)

   Mfano Mtoto anacheza vizuri (Alama $\frac{1}{4}$)

   (b) Alama $2\frac{1}{2}$

   Mimi na wewe tutadumu milele daima duniani.





Mimi na wewe tutadumu milele daima duniani. 1/2

6. (i) Kanga 1% (ii) Beberu 1% au Kupe (iii) Pikipiki au Ndala,
Bomu, zeze. 1% (iv) Ndege au mkono wa ndege. (v) Kifaru. 1%

7. (a) Ni muungano wa konsonanti na irabu au irabu na irabu inayotamkwa mara
moja kwa pamoja kama fungu moja la sauti. 02%.
Silabi wazi (pekee) inaishia na irabu 1%- Bao
Silabi funge inaishia na konsonanti 01%
(b) Ni sehemu ndogo ya neno isiyoweza kugawanyika zaidi. 01%

SEHEMU C

8. (a) Mwisho wa sentensi. 01%
Kuandika vifupisho vya maneno. 01% k.n.y S.L.B
(b) Kuonesha pumziko dogo. 01%
Kuandika anuani 01% Mtaa wa Kilimani,
S.L.P 50,
AU
Kutenganisha mwezi na mwaka
(c) - Hutumika kuonesha herufi au namba katika orodha. 1%
- Kufunga maelezo yanayofanana maana ya ziada ya neon. 1%
(d) Kuendeleza sauti mfano. 1% Mo-O-O.
Kuonesha silabi za neon mfano ki-ta-bu.
(e) Kuonesha orodha ya vitu. 01%
Katika uandishi wa vichwa vya habari. 1%

WIZI SHULENI

Mhariri
Gazeti la majira (sio lazima liwe majira hata gazeti lolote)
Ndugu Mhariri (Utangulizi wa jumla)
Ndugu mhariri aeleze jambo kwa kina 01
Ndugu mhariri ni matumaini yetu suala letu litashughulikiwa 01
Kushokashiza
S.L.P 10
IRINGA. Anuani na jina lolote

9. Mwanafunzi anatakiwa aandike kumbukumbu ya kikao kwa kuzingatia mambo yafuatayo

i. Kichwa cha habari kizingatiwe
- Sehemu itayofanyika kikao
- Siku, na tarehe
- J ina la kikao
- Muda / saa

ii. Mahudhurio
Lakini ainisha
- Waliohudhuria
- Wasiohudhuria

iii. Ajenda
Ajenda imegawanyika katika mambo manne
- Kufungua kikao
- Maandalizi
- Mengineyo
- Kufunga kikao

Mfano

MUKTASARI WA KIKAO CHA KAMATI YA WANAFUNZI KUANDAA SHEREHE YA KUMUAGA MKUU WA SHULE KILICHOFANYIKA 16/06/2019 KATIKA UKUMBI KUANZIA SAA 5:00 ASUBUHI MPAKA SAA 7:00 MCHANA (ALAMA 02)

WALIOHUDHURIA

i.	Pili jumapili	- Mjumbe	
ii.	Miriam	- Mjumbe	(Alama 1)
iii.	Enock	- Mjumbe	
iv.	Hiza Bashiri	- Mwenyekiti	
v.	Mudhihiri Harona	- Katibu	

WASIOHUDHURIA

i.	YUSNASRI YUSUPH	(bila taarifa)	(Alama 1)
ii.	ASHA OMARI	(kasafiri)	(Alama 1)

AJ ENDA

i.	Kufungua kikao	(Alama $1\frac{1}{2}$)
ii.	Maandalizi	(Alama 1)
iii.	Mengineyo	(Alama 1)
iv.	Kufunga kikao	(Alama $1\frac{1}{2}$)

i. KUFUNGUA KIKAO

Mwenyekiti alifungua kikao mnamo saa tano kwa kuwakaribisha wajumbe na

kuwaomba wawe huru kuchangia katika kikao hicho.

ii. MAANDALIZI

Ili kufanikisha sherehe hiyo wajumbe waliamua kuchagua kamati ndogo ambayo ingesaidia katika kufanikisha sherehe. Wajumbe wa kamati wlio chaguliwa
Ni i.) Pili J umapili
ii.) Enock
iii.)Miriam

iii. MENGINEYO

Wajumbe walikubaliana kuchangia kiasi cha shilingi 1,000 ili kumnunulia Zawadi mkuu wa pia walikubaliana katika kikao watu wajitokeze kwa wingi katika sherehe ya shule kumuaga mkuu wa shule.

iv. KUFUNGA KIKAO

Mwenyekiti alifunga kikao kwa kuwaomb wajumbe wawafikishie taarifa wanafunzi wote kuhusu yaliyokubaliwa katika kikao na kutoa ushirikiano kwa wingi ili kufanikisha sherehe hiyo.kikao kilifungwa saa 7:00 mchana.

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J umanne Bunju Hasara faida

Mwenyekiti (Alama $^{1}/_{2}$) katibu (Alama $^{1}/_{2}$)

16/05/2019 16/05/2019

(SEHEMU D ALAMA 10)
MAENDELEO YA KISWAHILI

10. UREJ ESHI. 02%
Vitenzi vya lugha ya kibantu vina tabia ya kukubali urejeshi na Kiswahili nacho vina tabia ya kukubali urejeshi.

Mfano: KISWAHILI	UREJ ESHI
Kuchoka	aliyechoka
Kigogo – kukatala	yakafaye
Kiha – ukuruka	uwurushe

UKANUSHI. 02%
Vitenzi vya kibantu na vya Kiswahili vyote vina tabia ya ukanushi.

Mfano:		UKANUSHI
KISWAHILI	Alicheka	Hakucheka
Kingindo	aliekite	ahekiteje

MWANZO WA VITENZI. 02%
Vitenzi vya lugha ya Kiswahili huanza na viwakilishi nafsi viambata na kibantu hivyo hivyo.

Mfano: KISWAHILI	KITENZI	KIAMBISHI NAFSI

Anasoma A –
Kizigua Aasoma A –

VIAMBISHI. 02%
Vitenzi vya kibantu na kugha ya Kiswahili vina viambishi awali,mzizi na viambishi tamati.
Anasoma – A- NA- SOM- A
AASOMA – A- A- SOM- A

MIISHO YA VITENZI. 02%
Vitenzi vya lugha ya Kiswahili na vya lugha ya kibantu huishia na irabu.

		IRABU
KISWAHILI	KUIMBA	- A
KIGANDA	OKUIMBA	- A
KIJITA	OKWIMBA	- A

N.B: Hakuna alama ya utangulizi wala hitimisho.

SEHEMU E

11. Utangulizi: Maana ya methali, fani. 02%
Muundo: Methali imegawanyika katika sehemu kuu mbili moja inaanzisha wazo na sehemu ya pili inakamilisha au inakanusha.03%
Mfano: Haraka haraka, haina Baraka
Polepole, ndio mwendo.
MTINDO: Methali inatumia mitindo mbalimbali.
- Uulizaji
  Pilipili usiyoila yakuwashia nini?
  AUngurumapo samba, mcheza nani 1.5
- Mtindo wa ushairi.
  Haraka haraka, haina Baraka (6,6)
  Mla nawe hafi nawe, bali mzaliwa nawe (8,8) ½

WAHUSIKA:
Wahusika wa methali wanaweza kuwa binadamu, wanyama, ndege n.k
Kunguru mwoga hukimbiza bawa lake.
Mbuzi wa maskini hazai.

MUKTADHA
Methali inaweza ikaonesha muktadha wa huzuni, kejeli au furaha. 03%
Ganda la miwa la jana chungu kaona kivuno.
Mie nyumba ya udongo sihimili vishindo.
N.B: Matumizi ya Lugha.
HITIMISHO

12. UTANGULIZI
Mwanafunzi anatakiwa atoe maana ya maneno yaliyotumika katika swali
Taswira

ni lugha ya picha itumikayo kueleza dhana mbalimbali katika kazi za fasihi.

Dhamira

Ni wazo lililo msukuma msani kuondoa kazi ya fasihi

Mbwakoko - (Alama 1.5)

Pia aonesha vitabu atakavyo vitumia katika swali hili sambamba na hilo aonyesha kupinga kiini

Mwanafunzi anatakiwa atoe hoja sita(6) tatu kutoka kila kitabu katika uandishi wa hoja azingatie

Taswira – (Alama1)

Maana ya hiyo taswira – (Alama $^{1}/_{4}$)

Shairi lililota hiyo Taswira- (Alama $^{1}/_{4}$) Alama 12

Dhamira iliyoibuliwa na hiyo Taswira - Alama $^{1}/_{4}$)

Ahustanishe na jamii - Alama $^{1}/_{4}$)

MALENGA WAPYA

Samaki – watu wa tabaka la chini – shairi la samaki mtungini

Dhamira **–** unyonyaji

Sukari – madaraka – shairi la sukari

Dhamira **–** uongozi mbaya

Ua - mwanamke- shairi la ua

Dhamira **–** mapenzi

CHEKACHEKA

WASAKATONGE

- Nahodha – kiongozi- katika shairi la nahodha – dhamira uongozi mbaya
- wasakatonge – watu wa tabaka la chini- shairi la wasakatonge dhamira – unyonyaji
- vinyonga- wasaliti - katika shairi la vinyonga – dhamira usaliti na unafiki.

Hitimisho

Mwanafun(Alama 1.5) Alama 9

13. UTANGULIZI: Maana ya mandhari, mgogoro.

WATOTO WA MAMA NTILIE

- Mandhari ya jaani yanatoa mgogoro kati ya peter & doto.
- Mandharin ya nyumbani kwa mama ntilie yanatoa mgogoro kati ya lomolomo na mkewe.
- Mandhari ya shuleni yanaibua mivutano kati ya Mwl. Chikoyo na peter, zita na musa.

TAKADINI.

- Nyumbani kwa sekai yanaibua mgogoro sekai na mumewe.
- Kijijini yanaibua mgogoro kati ya wanakijiji na sekai.

- Mkutano yanaibua mgogoro kati ya mtemi masasa na wanakijiji.

HITIMISHO 01%

14. UTANGULIZI: Watahiniwa wataonesha namna jina la kitabu linavyoundwa na ataeleza maana ya kusadifu dhamira.
NGOSWE – PENZI KITOVU CHA UZEMBE.
Maana yake mhusika mkuu ngoswe – Penzi ndio kiini cha Uzembe. Mtahiniwa ataonesha dhamira zilizojitokeza kutokana na jina hili.
    - UZEMBE.
    - MAPENZI.
    - ULEVI.

KILIO CHETU.
Maana yake kilio-kitendo cha kutoa machozi, chetu ni vijana na rika lote kuhusu dhamira zifuatazo.
    - UGONJWA WA UKIMWI.
    - ELIMU YA JINSIA.
    - MMONYOKO WA UDOGO.

HITIMISHO 01%

15. (a)Mwanafunzi anapaswa kutoa tofauti kati ya ngonjera na majigambo.
Ngonjera- ni shairi lenye pande mbili zinazojibizana
Kwa mfano ubeti wa kwanza Mwalimu - ubeti wa pili Mwanafunzi.
Majigambo – ni sanaa ya unenaji yenye utendaji na inayotumiwa na wasanii kujigamba ama kujinaki na kutangaza sifa zao. (Alama 1)

(b) Mwanafunzi atunge ngojera kwa kuzingatia kanuni zifuatazo:-
Kichwa
Wahusia wanojibizana
Urari wa vina na mizani (ziwe katika mshororo mmoja) na atunge (Alama1)

Mfano wa ngonjera
MFUMO DUME KATIKA JAMII

Mwanamke: Dunia yaenda wapi, usawa unapotea
Haki zetu sie zipi, ambazo mwatupatia
Hizi mmetoa wapi?, nyangumi mwatuonea (Alama3)
Hatuna sisi kauli, wanawake tunalia.

Mwanaume: Hasa mwataka nini, katika hii dunia
Kazi yenu ni jikoni, shambani na za kufua
Ya nini yenu maoni, kwa vidume kuyatoa (Alama3)
Mwanaume kila kitu, katika hii dunia.

Wote: Ukweli tumetambua, umoja kushirikiana
Sote ni sawa sikia, hakuna kujibagua (Alama 3)
Uonevu hilo doa, katika hii dunia
Tusitengane jamani, kuleta maendeleo.

BISMILLAH RAHMAN RAHIIM

THE NATIONAL MUSLIM COUNCIL OF TANZANIA
ZAIM EDUCATION DEVELOPMENT

FORM FOUR ZAIM SCHOOLS MOCK EXAMINATION

PHYSICS 1
MARKING SCHEME
(For Both School and Private Candidates)

1.

i	ii	Iii	iv	v	vi	vii	viii	ix	x
D	C	B	E	D	C	A	C	C	D

10 marks @ 01 mark

2.

i	ii	Iii	iv	v	vi	vii	viii	ix	x
C	M	E	G	A	H	J	B	K	I

10 marks @ 01 mark

3. **10 marks @ 01 marks**

   i. Hygrometer

   ii. Compressions

   iii. Very law

   iv. Magnetic field

   v. Pulleys

   vi. Mass, force

   vii. Specific heat capacity

   viii. Refractive index

   ix. Nuclear fission

   x. Metal rectifier

4. (a)(i) Moment of a force : Is the product of a force and perpendicular distance. It's SI

unit is (NM) **(01 marks)**

(II) Moment of a force is increased due to increase in distance from its line action from the fulcrum. **(01 marks)**

Examples:

- It is easier to push a door open the further you are from its hinges.
- It is easier to turn jack using a long Tommy bar than using a short one.

($^{01}/_{2}$ marks)

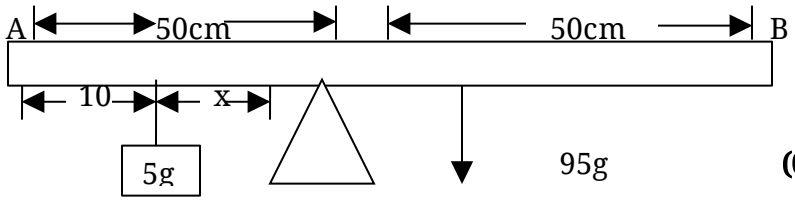
(b)(i) (a) The sum of clockwise moment must be equal to the sum of anticlockwise moment.
($^{01}/_{2}$ marks)

(b) Upward forces must be equal to the downward forces. **($^{01}/_{2}$ marks)**

(ii) The handle of a door is near the outside edge in order to increase the turning effect of the force applied to open or close the door
($01^{1}/_{2}$ marks)

(c)(i) Centre of gravity is the point where by total weight of a body is appear to be concentrated. **(01 mark)**

(ii)



(01 mark)

From, sum of clockwise moment = sum of unticlockwise moment

But

Distance btw x and y = 40cm

x + y = 40cm. **($^{01}/_{2}$ marks)**

5 x x = 95y. **($^{01}/_{2}$ marks)**

But x + y = 40

x = 40 – y

5x (40 – y) = 95y. **($^{01}/_{2}$ marks)**

200 – 5y = 95.

200 = 100y

y = 2

x + y = 40

But y = 2

x = 38 **($^{01}/_{2}$ marks)**

- ❖ The distance of knife – edge from the centre from lath is 2cm. **(01 mark)**

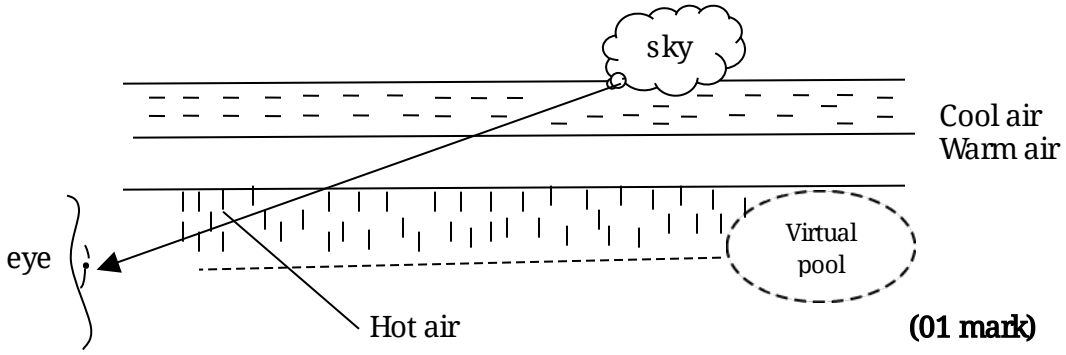
5. (a)(i) - Principal focus. Is the point where by a ray of light is close to and parallel to the principal axis strikes a concave mirror **($^{01}/_{2}$**

marks)

- Focal length. Is the distance from the pole to the principal focus. **($^{01}/_{2}$ marks)**

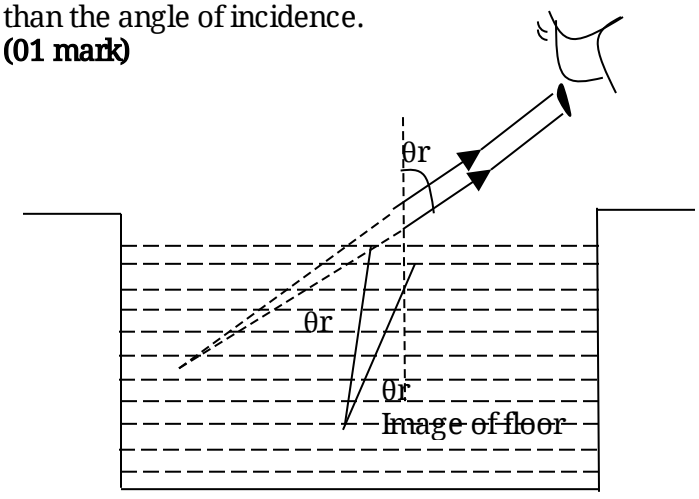
(ii) Mirage. Is a virtual pool of water observed at a distance on the surface on the road way, desert or an open land during a hot day.

- It's an image of the sky produced by progressive banding of light as is passes through the warmer air layers of decreasing refractive index until if become parallel to the ground and after which if proceed to bend upward.



(01 mark)

(iii) Because a light ray bends as its passes from water to air as rays if light from the bottom pass from water to air there are reflected at an angle of refraction greater than the angle of incidence.
(01 mark)



(01 mark)

(b)(i) Dispersion of white light. Is the spitting of white light into it's component colours when it pasess through a prisim.
($^{01}/_{2}$ marks)

(ii) - Primary color. Is the color that can not be created by mixing other colors.
- Complimentary color Are colors which when mixed in a definite ratio yield white. **($^{01}/_{2}$ marks)**

(c)(i) $M = \frac{V}{U}$ ———— (i)

$\frac{I}{F} = \frac{I}{U} + \frac{I}{V}$ ———— (ii)

Then.

$M = \frac{V}{U}$

$V = Mu$ **($^{01}/_{2}$ marks)**

From; $\frac{I}{F} = \frac{I}{U} + \frac{I}{V}$ **($^{01}/_{2}$ marks)**

But $V = Mu$

$\frac{I}{F} = \frac{I}{U} + \frac{I}{Mu}$

$\frac{I}{F} = \frac{M+1}{Mu}$ **($^{01}/_{2}$ marks)**

$F(\frac{M+1}{M}) = \frac{Mu}{M}$

$U = (\frac{M+1}{M})F$ **($^{01}/_{2}$ marks)**

Hence shown.

(c)(ii) Solution

$M = \frac{V}{U} = 4$

$V = 4U$ **($^{01}/_{2}$ marks)**

Thus xcm = U, 4xcm = V

- $V = -4x, U = +x$

From. $\frac{1}{V} + \frac{1}{U} = \frac{1}{F}$ **($^{01}/_{2}$ marks)**

. $\frac{1}{-4x} + \frac{1}{+x} = \frac{1}{(+20)}$ **($^{01}/_{2}$ marks)**

- $\frac{-1}{4x} + \frac{1}{x} = \frac{1}{20}$

- $\frac{-3}{4x} = \frac{1}{20}$

$4x = 60$ or $x = 15$ **($^{01}/_{2}$ marks)**

- Object distance = 15cm

6. (a)(i)- Interference. Is the phenomenon which occurs when two waves that are travelling along a same medium meet.

($^{01}/_{2}$ marks)

- Diffraction. Is the phenomenon which occurs when a wave encounters an abstacles.

($^{01}/_{2}$ marks)

(ii)- The frequency or pitch of a note from a bowed violin dring depends on the length of the string, it's tension, its Mass per Unit length or thickness, and the material. The violinist change the pitch of the note played by varying the length of on of the strings, and strings of different thickness enable him or her to obtain greater range of pitch. (011/2 marks)

(b)(i) Echo. Is the reflected sound which is heard distinctly from the original sound. While amplitude. Is the maximum displacement of a wave from equilibrium position.

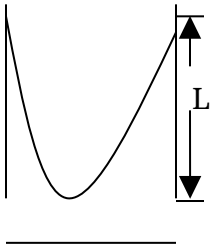
($^{01}/_{2}$ marks)

(ii) Sonometer. Is the instrument which used to study the properties of a wave especially stationary **($^{01}/_{2}$ marks)**

(c)(i) Note on the violin and flute have different quality because each fundamental note is accompanied by overtones and the overtone of a string instrument are different of those of a wind instrument. This gives the different wave forms which determine the quality of the not

($01^{1}/_{2}$ marks)

(ii)(a) solution



Data given.

Length = 100mm ≈ 10cm

But, length of pipe is $\lambda/4$ **($^{01}/_{2}$ marks)**

$\lambda/4$ = 10cm **($^{01}/_{2}$ marks)**

$\lambda/4$ = 4×10 = 40cm = 0.4M **($^{01}/_{2}$ marks)**

- The fundamental frequency $f_0 = \frac{v}{\lambda} = \frac{340}{0.4}$
- f_0 = 850Hz (01 mark)

(b) First overtone.

Frequency $f_0 = 3f_0 = 3\times850 = 2550HZ$ **($^{01}/_{2}$ marks)**

(c) If the temperature changes during the day, the velocity of sound V changes since V depends on the air temperature. So the frequency which is $\frac{v}{\lambda}$ changes.

($01^{1}/_{2}$ marks)

7. (a)(i) Momentum. Is the product of Mass and its velocity. **($^{01}/_{2}$ marks)**

(ii) Seat belts are allowed to stretch upon collision in order to increase the time the force exerted by the belt on the body act when this happen, for the same change in Momentum, the force experienced by a person due to the belt is reduced.

Impulse = Force ×Δtime = Δmomentum

$$\text{Force} = \frac{\Delta\text{momentum}}{\Delta\text{time}}$$

So, when a belt stretches, Δtime is increased and so the force on a person by the belts is reduced. **(02 marks)**

(b)(i) Inertia. Is the ability of a body at rest to resist motion **($^{01}/_{2}$ marks)**

(ii) From

Rate = 50 g/s.

= 0.05 kg/s

From

$F = \frac{Mv-Mu}{t}$ but u = 0 **($^{01}/_{2}$ marks)**

$F = \frac{Mv}{t}$ **($^{01}/_{2}$ marks)**

$F = \left(\frac{M}{t}\right)V$ but $^{M}/_{t}$ = 0.05kg/s

F = 0.05kg/s. V. **($^{01}/_{2}$ marks)**

But

V = 400cm/s = 0.4m/s.

F = 0.05kg/s. X 0.4m/s

= 0.02N. **(01 mark)**

(c)(i) From the law of conservation for A and B Final total momentum = Initial momentum **($^{01}/_{2}$ marks)**

Final total momentum A and B

= 1Mass x Velocity of B) – (Mass x Velocity of A)

3V – (2 x 60). **($^{01}/_{2}$ marks)**

Initial total momentum of A and B = 0

3V – 120 = 0 **($^{01}/_{2}$ marks)**

From which V = 40m/s **($^{01}/_{2}$ marks)**

(ii) Total kinetic energy of A and B

$= \left(^{1}/_{2}\times 2\times 60^{2}\right)+\left(^{1}/_{2}\times 3\times 40^{2}\right)$ **($^{01}/_{2}$ marks)**

$= \left(\frac{1}{2} \times 2 \times 3600\right) + \left(\frac{1}{2} \times 3 \times 1600\right)$ **(01/2 marks)**

$= 3600 + 2400 = 600J$ **(01 mark)**

8. (a)(i) Nuclear fusion. Is the process in which lighter nuclei join together to form a heavier nucleus. **(01 mark)**

Example:

$${}^{2}_{1}H + {}^{3}_{1}H \longrightarrow {}^{2}_{2}He + {}^{1}_{0}n + \text{Energy,}$$ **(01/2 marks)**

(ii) Data given

T = 168 seconds

$\frac{N_n}{N_o} = \frac{1}{8}$ **(01/2 marks)**

$t_{1/2} = ?$

From

$\left(\frac{1}{8}\right) = \left(\frac{1}{2}\right)^n$ **(01/2 marks)**

$\left(\frac{1}{2}\right)^3 = \left(\frac{1}{2}\right)^n$ **(01/2 marks)**

$n = 3$

but

$n = \frac{T}{t_{1/2}}$ **(01/2 marks)**

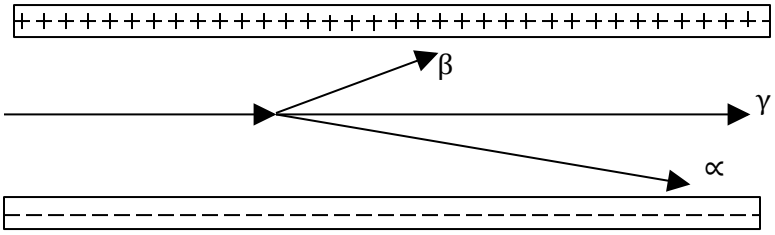
❖ $t_{1/2} = \frac{T}{n}$ **(01/2 marks)**

$= \frac{168}{3} = 56$ seconds **(01/2 marks)**

❖ This means that if takes 56 seconds for a half of lhoron to disintegrate.

- Half life. Is the time required for one half of the nuclei to decay. **(01/2 marks)**

(b)(i)



(01 1/2 marks)

(ii) 1) $^{285}_{92}M \longrightarrow {}^{284}_{90}X + {}^{4}_{2}H$ ($\propto$ δ - particle) **(01/2 marks)**

2) $^{284}_{90}X \longrightarrow {}^{284}_{11}Y + {}^{0}_{-1}e$ (δ - particle) **(01/2 marks)**

3) $^{284}_{11}Y \longrightarrow {}^{284}_{11}A + \delta$ (δ - particle) **(01/2 marks)**

(c)(i) Gastrointestinal track, proliferating cells in the mucus membrane that lines the track are easily killed by irradiation, resulting in the denudation and ulceration of the mucus membrane.
(01 mark)

(ii) Reproductive organs, rapid exposure of both testes to even a dose may interrupt sperm production temporary. A high-dosage exposure may cause permanent sterility in men.
(01 Mark)

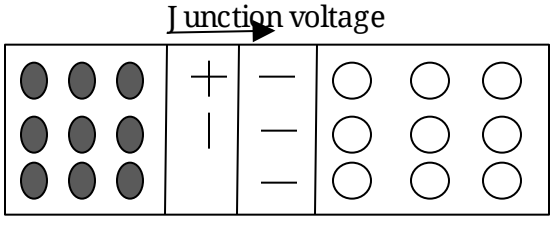
(iii) Eyes irradiation can cause opasification of the lens and severity of which increase with the dose.
(01/2 marks)

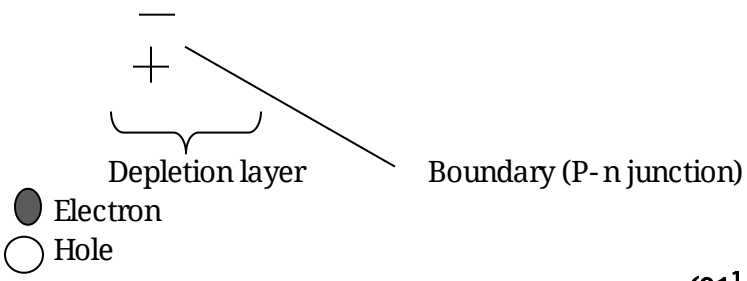
9. (a)(i) P-type semiconductors are produced from addition of trivalent atoms into the semiconductor lattice. Such atoms lead to formation of three covalent bonds with semiconductor atom leaving a shortage of one electron from the normal silicon electronic structure. This missing electron behaves as a positive charge carrier since as it is filled by neighboring electron the vacancy shifts in the opposite direction.
(01 mark)

(ii) N-type semiconductor. Are produced once the group V atom is incorporated into the crystals lattice it makes four covalent bond with semiconductor atoms leaving one of its electrons only weakly bound. Electron so created is available for conduction even at normal temperature.
(01 mark)

(iii) Transistor can be formed as either an individual component or as part of a continuous component called integrated circuit.
(01 mark)

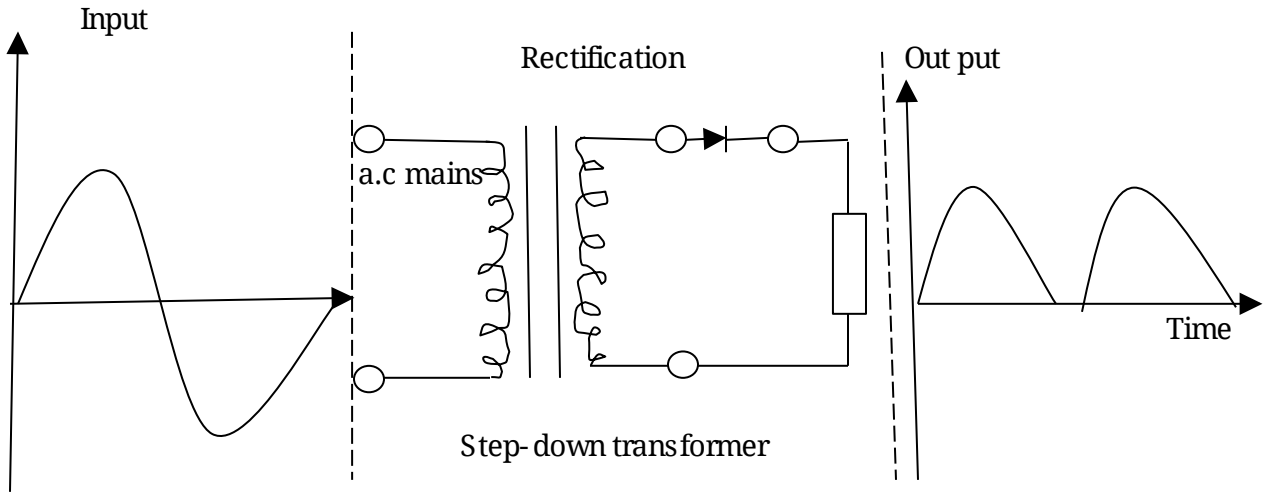
(b)(i) As soon as a p-n junction is made electrons cross over the boundary from the n-type material to holes in the P-type material. At the same time holes cross over from the P-side and capture electrons. This movement of holes and electrons cause the n-side to become positively charged and the P-side negatively charged, thus, creating a potential difference across the boundary.





(01 1/2 marks)

(ii) Diodes in half-wave rectification of a.c current allows current only during the of the cycle when its forward biased and block current when it is in reverse direction.



(01 1/2 marks)

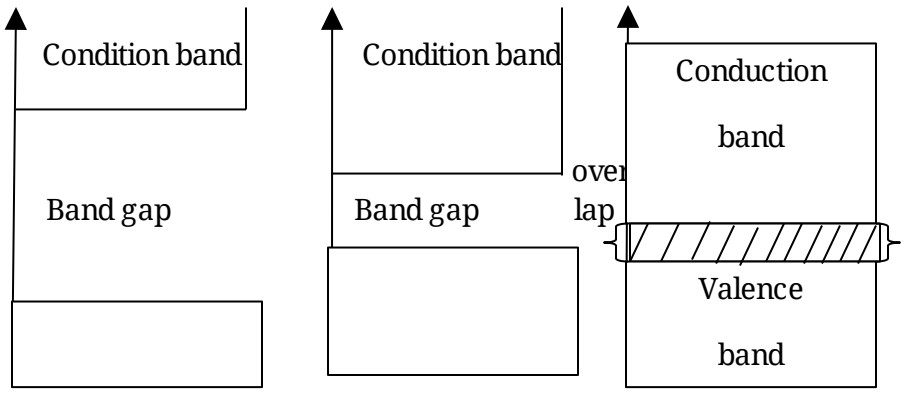
(c)(i) Conductor. Are materials that contain mobile electric charges **(01/2 marks)**

(ii) Semiconductor. Is a material with an electrical conductance that is intermediate between that of an insulator and conductor. **(01/2 marks)**

(iii) Insulator. Is a material or object which resist the flow of electrical charges through it.
(01/2 marks)

(d) The conduction band is range of electron energy higher than the valence band. In metals the valence and conduction band overlap. The valence band is only partially. In conductors the valence and conduction band overlap hence electrons are free to participate in conduction of an electric current. For insulator the for bidden gap is large so at low temperature no electron re available for electron. In a semi conductor band gap is medium in size so election may jump from valence band to conduction band. at high temperature but at low

temperature may not.



Insulator Semiconductor Conductor

(02$^{1}/_{2}$ marks)

10. (a)(i) Star. Is the large celestial body made up of hot gases known as plasma **($^{01}/_{2}$ marks)**

(ii) Planet. Is the major object which is in orbit around a star. **($^{01}/_{2}$ marks)**

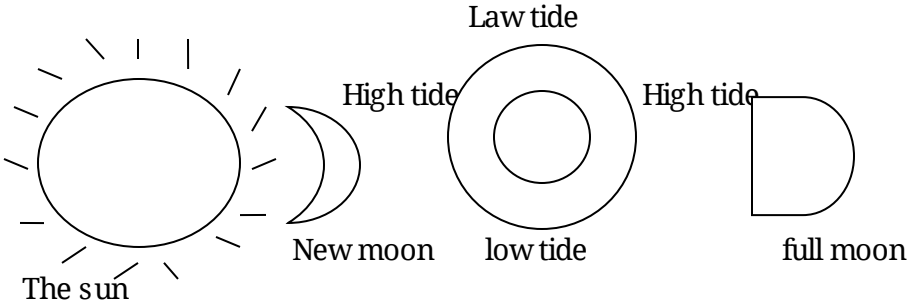
(iii) Consellation. Is the group of stars hat form a definite shape or pallern when viewed from the earth.
($^{01}/_{2}$ marks)

(b)(i) (a) Are used in religious activities **($^{01}/_{2}$ marks)**

(b) Are applied in agricultural activities. **($^{01}/_{2}$ marks)**

(c) Also used in navigation **($^{01}/_{2}$ marks)**

(ii) How tides are formed.
The moon's gravitational pull tries to pull ever thing around it including the earth' but since the earth is heavy the moon only pulls the waters it to a point where the moon's pulls balances the earth's pull on the waters when this happens water on that side bulges towards the moon. Since the earth rotates on its axis two tides occur each day.



(01$^{1}/_{2}$ marks)

(c)(i) Earth quake. Is a sudden motion or shaking of the earth caused by sudden release of energy that has accumulated with or along the edges of the earth's tectonic plates.

(01 mark)

(ii) Precaution **(mark only three)**

- If you are inside take cover under a table or desk and stay away from falling objects
- If you outside stay far from buildings. Power lines, street light etc.
- Stay in one place until shaking stops
- Turn off power (main switch) to avoid electric shocks.
- Be on the lookout for fire.

(011/2 marks @ 01/2 marks)

(iii) Effect of volcanoes. **(mark with explanation only two)**

- Landscape formation
- Volcanic eruptions
- Pollution
- Destruction of property and loss of human life.

(02 marks @ 01 mark)

(d)(i) Hypocentre. Is a point within the earth where an earth quake begins **($^{01}/_{2}$ marks)**

(ii) Epicenter. Is a point on the earth's surface directly above the hypocentre. **($^{01}/_{2}$ marks)**

11. (a)(i) Law of flotation states; that a floating body displace its own weight of a fluid in which it float. **($^{01}/_{2}$ marks)**

(ii) Ship float but the piece of iron sink because, a ship has a large volume hence exert a low density than the density than the density of water therefore it float, while a piece of iron has a smaller volume hence exert a higher density than the density of water.**(01$^{1}/_{2}$ marks)**

(b)(i) Condition for a body to float

a) Up thrust must be greater than the weight of a body. **(01 mark)**

b) The density of object must less than the density of fluid **(01 mark)**

(ii) Solution

Mass of water displaced = 28g.

Volume of water displaced = 28cm^{3}.

Assuming top of stem to be flat and not rounded.

Therefore Volume of stem above = $0.75 \times 3 = 2.25cm^3$
Total volume of hydrometer = $28 \times 2.25 = 30.25cm^3$ **01 mark**

Volume of liquid displaced = $\frac{\text{mass}}{\text{density}}$

$= \frac{28}{1.4} = 20cm^3$ **($^{01}/_{2}$ marks)**

Therefore volume of stem above the liquid = $30.25 - 20$

$= 10.25cm^3$ **($^{01}/_{2}$ marks)**

Hence length of stem above liquid

$\frac{10.25}{0.75} = 13.7cm$. **($^{01}/_{2}$ marks)**

(c)(i) Up thrust. Are upward force acting normally on an object when body is totally or partially immersed in fluid. **($^{01}/_{2}$ marks)**

(ii) solution

Data given.

Mass of sea-water displaced = 1200t

Density of sea water = $1000 \times 1.03kg/m^3$.

Volume displaced = $\frac{\text{mass}}{\text{density}} = \frac{1200 \times 1000}{1000 \times 1.03}m^3$

$= 1165m^3$ of sea water. **(01 mark)**

The same volume of fresh water has a mass of $1165 \times 1000kg = 1165t$. **(01 mark)**

Therefore, Mass of cargo to be unloaded

$= 1200 - 1165 = 35t$. **(01 mark)**